

m-format constellation -- Command-line Examples

mnemonic-toolkit v0.55.3 -- worked examples (Linux), exact verbatim I/O

2026-06-15

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About these examples

This document shows real, copy-pasteable command lines for the **m-format constellation** -- a steel-engravable Bitcoin backup system built around four CLIs (`mnemonic`, `md`, `ms`, `mk`). Every command below was executed against `mnemonic v0.55.3` on Linux and **both the command and its full output are reproduced verbatim** -- no abbreviations, no ellipses, no elided keys or addresses. Long lines wrap with a grey hook-arrow continuation marker in the left margin.

All seed phrases here are public BIP-39 TEST VECTORS (`abandon abandon ... about`, etc.). They are world-known and hold no funds. **Never type a real seed phrase onto a networked machine, and never reuse these test phrases for real money.** The toolkit is alpha software -- use only with disposable amounts or on testnet until it has been independently audited.

Seed input from a file

Yes -- the toolkit reads a seed phrase from a file. There is no `--phrase-file` flag; instead every secret slot accepts the value `-`, meaning **"read this secret from stdin"**, which you point at a file with the shell's `< file` redirect:

```
$ mnemonic bundle --template bip84 --network mainnet --slot @0.phrase=- < seed.txt
```

This is the **secure** idiom: the phrase travels on stdin, so it never appears in `argv` / `/proc/$PID/cmdline` or your shell history (unlike the inline form `--slot @0.phrase='<words>'`, which the toolkit flags as `secret-on-argv`). The reader strips a single trailing newline, so an ordinary one-line text file works.

Multiple files for multiple seeds

Only **one** stdin secret is allowed per invocation, so `< file` reads exactly one seed. To use several seed files there are two patterns:

1. **One file per invocation (secure -- recommended).** Process each seed file in its own command and combine only the resulting public xpubs. No machine ever holds more than one seed, and nothing secret reaches `argv`. This is the per-device 2-of-3 multisig flow in section 3.
2. **Several files in one command (convenient -- less safe).** Read each file with command substitution, `--slot "@N.phrase=$(cat seedN.txt)"`. This works for any number of seeds, but each substituted phrase lands on `argv` (the toolkit prints a `secret-on-argv` warning). Only one slot may use the `-` stdin form; a second `-` is rejected. Shown in section 3.4.

Convention in this document: whenever a command reads a file (a seed, a descriptor, a policy JSON, or an `md1` chunk list), the file's contents are printed with `cat` immediately beforehand, so every input is visible.

The three engraved cards:

Card	What it carries
ms1	BIP-39 entropy (recovers the seed)
mk1	xpub + origin (master fingerprint + BIP path)
md1	wallet policy (descriptor template + bound xpubs)

Throughout, `$` is the shell prompt; everything after it is what you type.

1. Install the constellation on Linux

The in-repo installer builds each component with `cargo install --locked` into `~/cargo/bin` (no sudo, no system files touched). It needs cargo, git, and a C toolchain; the CLIs require `rustc >= 1.85`.

Install all four CLIs (this compiles from source, so the build log is machine-specific and not reproduced here):

```
$ sh -c "$(curl -fsSL https://raw.githubusercontent.com/bg002h/mnemonic-toolkit/master/scripts/install.sh)" -- --no-gui
```

The installer carries the current version pins, so it never goes stale. Useful flags: `--only <c>`, `--exclude <c>`, `--no-gui`, `--from-git`, `--force`, `--dry-run`, `--list`. The pin table (`--list`) and a dry run are deterministic:

```
$ sh '/scratch/code/shibboleth/mnemonic-toolkit/scripts/install.sh' --list
```

COMPONENT	CARGO_PACKAGE	DEFAULT	FEATURES	GIT_TAG
-----	-----	-----	-----	-----
mnemonic	mnemonic-toolkit	git (only)	(none)	mnemonic-toolkit-v0.56.0
md	md-cli	crates.io	cli-compiler	descriptor-mnemonic-md-cli-v0.6.2
ms	ms-cli	crates.io	(none)	ms-cli-v0.7.0
mk	mk-cli	crates.io	(none)	mk-cli-v0.8.0
mnemonic-gui	mnemonic-gui	git (only)	(none)	mnemonic-gui-v0.40.0

```
$ sh '/scratch/code/shibboleth/mnemonic-toolkit/scripts/install.sh' --no-gui --dry-run
m-format constellation installer
install root: /home/bcg/.cargo/bin
source: crates.io (default; mnemonic-toolkit stays on git+tag)

install mnemonic (git: mnemonic-toolkit-v0.56.0)
[dry-run] cargo install --locked --git https://github.com/bg002h/mnemonic-toolkit --tag mnemonic-toolkit-v0.56.0 mnemonic-toolkit
install md (crates.io: md-cli)
[dry-run] cargo install --locked --features cli-compiler md-cli
install ms (crates.io: ms-cli)
[dry-run] cargo install --locked ms-cli
install mk (crates.io: mk-cli)
[dry-run] cargo install --locked mk-cli
skip mnemonic-gui

4 installed.

verify:
  mnemonic --version      md --version
  ms --version            mk --version
  mnemonic-gui --version
```

Verify the install and list every subcommand:

```
$ mnemonic --version
mnemonic 0.55.3
```

```
$ mnemonic --help
engraving-bundle CLI for the m-format star (msl + mk1 + md1)

Usage: mnemonic [OPTIONS] <COMMAND>

Commands:
  bundle          emit a 3-card engraving bundle from a phrase or xpub
  verify-bundle   round-trip-check an engraved bundle
  convert         convert between seed/key formats (BIP-39 / BIP-32 / WIF / msl / mk1)
  addresses       list an existing nostr key (npub/nsec) as Bitcoin addresses/descriptors/WIF
  decode-address  decode a Bitcoin address → network(s) / script type / witness version / scriptPubKey
  export-wallet   emit watch-only wallet artifacts (Bitcoin Core importdescriptors, BIP-388 wallet_policy)
  import-wallet   import a third-party wallet blob into an m-format bundle (v0.26.0 Phase 2: BSMS Round-2 only)
  derive-child    derive deterministic child entropy / keys from a master xprv (BIP-85)
  electrum-decrypt decrypt an Electrum field-encrypted secret (seed phrase / xprv) with a password
  final-word      emit the set of BIP-39 last words that yield a valid checksum for an N-1 partial phrase
  seed-xor       split a BIP-39 phrase into N XOR shares OR combine N shares back into a phrase
  seedqr         encode/decode SeedQR (BIP-39 mnemonic ↔ numeric digit-string QR payload)
  nostr           Wrap an existing nostr key (npub/nsec) as Bitcoin addresses/descriptors/WIF
  silent-payment  Derive a BIP-352 silent-payment receiver address (base + labeled) from a seed
  slip39         split a master secret into SLIP-39 K-of-N shares OR combine shares back (Trezor-compatible)
  ms-shares      split a secret into BIP-93 codex32 K-of-N (msl) shares OR combine shares back
  gui-schema     emit SPEC §7 GUI-overlay flag-surface schema JSON (companion to `mnemonic-gui` v0.2)
  repair         BCH error-correct a corrupted m-format card (msl / mk1 / md1)
  inspect        describe the contents of an m-format card (msl / mk1 / md1)
  compare-cost   compare wsh-vs-tr per-spending-condition cost for a miniscript or descriptor
  xpub-search    search for a target (xpub, descriptor, address, or passphrase) under a seed or xpub
  verify-message verify a Bitcoin message signature (legacy P2PKH signmessage + BIP-322 segwit/taproot)
  restore        emit a watch-only restore document (single-sig) from a seed + optional passphrase
  build-descriptor build a validated wsh(...) descriptor + BIP-388 policy from a JSON policy-tree spec
  help           Print this message or the help of the given subcommand(s)
```

Options:

```
--no-auto-repair v0.22.0 – skip auto-fire repair on decode failures; preserve pre-v0.22 exit policy. Global flag. Honored by `convert`, `inspect`,  
↳ and (v0.22.1+) `verify-bundle`. For `verify-bundle`, auto-fire is additionally gated on `std::io::stdout().is_terminal()` to preserve the  
↳ legacy VerifyCheck-row behavior when output is piped or captured (per v0.22.1 D18 – TTY-conditional default). Standalone `repair` ignores this  
↳ flag (the whole point of that subcommand IS repair). Under `--json` calling contexts the auto-fire emits a structured JSON envelope on stdout ( per v0.22.1 D20) instead of text-form  
-h, --help          Print help  
-V, --version       Print version
```

RECOVERING A FORGOTTEN BIP-39 PASSPHRASE:

If you have your seed words (entropy) but not the BIP-39 passphrase (the optional "25th word"): if you have a LIST of likely passphrases, `mnemonic xpub-search passphrase-of-xpub --passphrase-candidates-file <file> --target-xpub <a known xpub>` tests each candidate against a value you already know. To GENERATE or mutate a keyspace (wordlists, masks, typo models), `mnemonic` does not – an external open-source tool does: `btcrecover` searches passphrase candidates and confirms each by deriving an address / xpub / master-fingerprint at common default paths and matching a value you already know.

`btcrecover` (maintained): <https://github.com/3rdIteration/btcrecover>
original: <https://github.com/gurnec/btcrecover>

Pointer current as of 2026-05-25. Run untrusted recovery tools offline, on an air-gapped machine.

2. Single-sig card set from a seed phrase (file input)

Create a native-segwit (BIP-84, $m/84'/0'/0'$) single-sig 3-card bundle from one seed phrase held in a file. Write the phrase to `seed0.txt` (here a public test vector) and feed it on stdin:

```
$ printf '%s\n' 'abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about' > seed0.txt
```

```
$ cat seed0.txt
abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about
```

(`--template` choices for single-sig: `bip44`, `bip49`, `bip84`, `bip86`.) Run the bundle. `stdout` carries the three cards to engrave; `stderr` carries the human-readable engraving panel and the secret-material warning:

```
$ mnemonic bundle --template bip84 --network mainnet --slot @0.phrase=- < seed0.txt
# ms1 (entropy, BCH-checksummed)
ms10entrsqqqqqqqqqqqqqqqqqqqqqqqqqqqqc9sraq34v7f

ms10e ntrsq qqqqq qqqqq qqqqq qqqqq qqqc9 sraq 34v7f

# mk1 (xpub + origin)
mk1qprsqhqqq3cqt3leutks2qvz3vs70mejkhk622ws2kgdemj2cd8zwj2skzx2wq0qw70l4q99vdyh5x0z8v4yslsp8qp3y3dpe854wq4
mk1qprsqhqp0f30mtxzd65mvwcur9usdatwuvq6z70r9nwrgrk6xn6l8gy6nwa2n977sw6zh34rma0nh

mk1qp rsqhp qqsq3 cqt3l eutk s2qvz g3vs7 0mej h k622w s2kgd
emj2c d8zwj 2skzx 2wq0q w70l4 q99vd yh5x0 z8v4y slsp8 qp3yx
g3dpe 854wq 4
mk1qp rsqhp p0f30 mtxzd 65mvw cur9u sdatw uvq6 z70r9 nwrgrk
6xn6l 8gy6n wa2n9 77sw6 zh34r ma0nh

# md1 (wallet policy)
md1fgdxlpqpqm6jzzqqvqpdw0za5zs4ggy55aq4vsmnh4s6wyaypu34c7raqu8np
md1fgdxlpqf2zcgefcpumel75q5435j7seugaj5jr7qyur6vt76es5cdeyrq7zdy0d
md1fgdxlpq3xa2dk8vwpj7gx74hwqxqdp083jehp5tdrf0n5zdfkqcdlrvnh5r62jn

md1fg-dxlpq-pqpm6-jzzqq-vqpdq-w0za5-zs4gy-y55aq-4vsmn-hy4s6-wyayp-u34c7-raqu8-np
md1fg-dxlpq-f2zcg-efcpu-pmel7-5q543-5j7se-ugaj5-jr7qy-ur6vt-76es5-cdeyr-q7zdy-0d
md1fg-dxlpq-3xa2d-k8vwp-j7gx7-4hwqx-qdp08-3jehp-5tdrf-a0n5z-dfkqc-dlrvn-h5r62-jn

# === Wallet bundle: bip84, mainnet ===
# ms1: 1c017
# mk1: 1c017
# fingerprint: 73c5da0a
# origin path: m/84'/0'/0'
# Template: bip84
# md1: 1c01
warning: stdout carries private key material (can spend) – redirect or encrypt (e.g. '> file.txt' or '| age -e ...')
```

Each card is printed twice: once unbroken, once grouped into 5-character blocks (`ms10e ntrsq qqqqq ...`) -- the grouped form is what you punch or engrave. Add `--no-engraving-card` to suppress the `stderr` panel when piping into other tools.

3. Conventional 2-of-3 multisig from 3 seed phrases (per-device, file input)

A real multisig never lets one machine see more than one seed. Each cosigner derives **only their public xpub** from their own seed file (on their own, ideally air-gapped, device); the coordinator then combines the three **public** keys into a watch-only `wsh(sortedmulti(...))`. No secret ever leaves its file.

Put each cosigner's seed in its own file:

```
$ printf '%s\n' 'abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about' > seed0.txt
```

```
$ printf '%s\n' 'legal winner thank year wave sausage worth useful legal winner thank yellow' > seed1.txt
```

```
$ printf '%s\n' 'letter advice cage absurd amount doctor acoustic avoid letter advice cage above' > seed2.txt
```

On each device, derive that cosigner's BIP-87 multisig fingerprint and account xpub (`--template wsh-sortedmulti` implies the `m/87'/0'/0'` path) from the seed file. Cosigner @0:

```
$ cat seed0.txt
abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about
```

```
$ mnemonic convert --from phrase=- --to fingerprint --template wsh-sortedmulti --network mainnet < seed0.txt
fingerprint: 73c5da0a
```

```
$ mnemonic convert --from phrase=- --to xpub --template wsh-sortedmulti --network mainnet < seed0.txt
xpub: xpub6DBjYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuCG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM
note: stdout is watch-only – public keys only, cannot spend
```

Cosigner @1:

```
$ cat seed1.txt
legal winner thank year wave sausage worth useful legal winner thank yellow
```

```
$ mnemonic convert --from phrase=- --to fingerprint --template wsh-sortedmulti --network mainnet < seed1.txt
fingerprint: b8688df1
```

```
$ mnemonic convert --from phrase=- --to xpub --template wsh-sortedmulti --network mainnet < seed1.txt
xpub: xpub6CbhrPzY2z7NcCGCGjLAJLq8iRyjUfwmDXQs66MxTVURekqb9DpLnVJ5D1qpatZjUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokYLVsyEFZFt9
note: stdout is watch-only – public keys only, cannot spend
```

Cosigner @2:

```
$ cat seed2.txt
letter advice cage absurd amount doctor acoustic avoid letter advice cage above
```

```
$ mnemonic convert --from phrase=- --to fingerprint --template wsh-sortedmulti --network mainnet < seed2.txt
fingerprint: 28645006
```

```
$ mnemonic convert --from phrase=- --to xpub --template wsh-sortedmulti --network mainnet < seed2.txt
xpub: xpub6DB7HNqW6CZojxN85NuFTPWZhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
note: stdout is watch-only – public keys only, cannot spend
```

Wrap each as an origin-annotated descriptor key `[fingerprint/87'/0'/0']xpub` and combine into a 2-of-3 sorted-multisig descriptor (`/<0;1>/*` = the external/change multipath). The assembled descriptor file:

```
$ cat multisig.desc
wsh(sortedmulti(2,[73c5da0a/87'/0'/0']xpub6DBjYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuCG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM
↳ /<0;1>/*,[b8688df1/87'/0'/0']xpub6CbhrPzY2z7NcCGCGjLAJLq8iRyjUfwmDXQs66MxTVURekqb9DpLnVJ5D1qpatZjUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokYLVsyEFZFt9
↳ /<0;1>/*,[28645006/87'/0'/0']xpub6DB7HNqW6CZojxN85NuFTPWZhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
↳ /<0;1>/*))
```

Validate and canonicalise it (this also computes the BIP-380 checksum):

```
$ mnemonic export-wallet --descriptor "$(cat multisig.desc)" --format descriptor --network mainnet
wsh(sortedmulti(2,[73c5da0a/87'/0'/0']xpub6DBj1Ync4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM
  ↳ /<0;1>/*,[b8688df1/87'/0'/0']xpub6CBhrPzY2z7NcCGCgJLAJLq8iRyJUfwmdXQs66MxTVURekqb9DpLnVJ5D1qpatZJUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokYLVsyEFZFt9
  ↳ /<0;1>/*,[28645006/87'/0'/0']xpub6DB7HNqw6CZojxN85NuFTPWZhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTFmFKKA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
  ↳ /<0;1>/*)4wup4at0
note: stdout is watch-only – public keys only, cannot spend
```

The first receive address (here via the BSMS / BIP-129 record, which also carries the /0/*,/1/* derivation):

```
$ mnemonic export-wallet --descriptor "$(cat multisig.desc)" --format bsms --network mainnet
BSMS 1.0
wsh(sortedmulti(2,[73c5da0a/87'/0'/0']xpub6DBj1Ync4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM
  ↳ /<0;1>/*,[b8688df1/87'/0'/0']xpub6CBhrPzY2z7NcCGCgJLAJLq8iRyJUfwmdXQs66MxTVURekqb9DpLnVJ5D1qpatZJUuPGTyxf5TYU1vA34YFE9FHB4TvfYmokYLVsyEFZFt9
  ↳ /<0;1>/*,[28645006/87'/0'/0']xpub6DB7HNqw6CZojxN85NuFTPWZhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTFmFKKA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
  ↳ /<0;1>/*)4wup4at0
/0/*,/1/*
bc1qksssl2m6t3aynza394sr9m86vt6md2v76kj52jun2xlrdeaa4q84qtpl
note: stdout is watch-only – public keys only, cannot spend
```

Engrave the shared watch-only card set from the public descriptor (the md1 policy card is shared by all cosigners; each cosigner additionally backs up their own seed as a single-sig ms1 set per section 2). With only public xpubs supplied, the ms1 cards are empty placeholders:

```
$ mnemonic bundle --descriptor-file multisig.desc --network mainnet
# ms1 (omitted – descriptor watch-only mode)

# mk1[0] (cosigner 0 xpub + origin)
mk1qperpupqqspu3s7denyv8nwverpumnrrnchd5pcy3zepa59349dcs5n5stvpvk3v8x4eqsdngy2jl4wddt5ac2ptv4fya76tpreapyrdfr
mk1qperpupppq8k6pccpqr7svxkwx5ag99s9zfyml9v7tcqrexmpdj7jgqgmny8rr0z7vlj7eqzv2486xkkcfrzq8

mk1qp erpup qqspu 3s7de nyv8n wverp umnrrn chd5 pcy3z epa59
349dc gs5n5 stvpv k3v8x 4eqsd ngy2j l4wdd t5ac2 ptv4f ya76t
preap yrdfq r
mk1qp erpup pnq8k p6xcp qphr7 svxwk xx5ag 99s9z fymL9 v7tcq
rexmp dj7jg qgmny 8rr0z 7vlj7 eqzv2 486xk kcftr zq8

# mk1[1] (cosigner 1 xpub + origin)
mk1qperpupqqspu3s7denyv8nwverpumn9cdxzlzpcy3zepaqxfl630m4as45q6fz4ltsntlues3e3gylcu6jsa6jd69hy0whcpg2f28j5awu
mk1qperpapp02p3qtffjqpxtu9j5jaevqs7jqje40324vxlw0txswawpxte3zmhzp6lrpj3ga2lw65h2dd24cuerkpa

mk1qp erpup qqspu 3s7de nyv8n wverp umn9c dzxzl pcy3z epaqx
afl63 0m4as 45q6f z4lts ntlue s3e3g ylcu6 jsa6j dz69h y0whc
pg2f2 8j5aw u
mk1qp erpup p02p3 qtffj qpxtu j95jz aevqs 7jqje 40324 vxlfw
0txsw awpxt e3zmh zp6lr pj3ga 2lw65 h2dd2 4cuer kpa

# mk1[2] (cosigner 2 xpub + origin)
mk1qperp7pqspu3s7denyv8nwverpumpnpvg3gqvcy3zapanecw97sn9g25uukrgxca47at8as54xhd0rkqg57cs78mc8fk6507cs8tugnnpnf
mk1qperp7ppr4e59ef7u5p038j900ye439hsphz506fra5k3eac5kg0n7jhn3kpzpn6n0lygd1pfeq4a75lc2mrrz7f

mk1qp erp7p qqspu 3s7de nyv8n wverp umnpv v3gqv pcy3z epa59
cw97s n9g25 uukrg xca47 at8as 54xhd 0rkqg 57cs7 8mc8f k6507
cs8tn ugnpn f
mk1qp erp7p pr4e5 9ef7u 5p038 j900y e439h sphez 506fr a5k3e
ac5kg 0n7jh n3kpz pna6n 0lygd lpfeq 4a75l c2mrr z7f

# md1 (multisig wallet policy)
md1f5przzspq3m67zzqqvzrs3pstucw0za5znwrg3hcc5xg5qxzhs7ygg2f6g9kqktgkrn2spp6tlfzprv6ye
md1f5przzswgyrv6p2shatnt2a8wzs2m92f2fsmqarvqsqm3lgr8trr2w5zjcz3yjdjk0qf2g9tn3u5jtsz
md1f5przzj7qq7fkctvh5jqxuepcmcchn8u30m4as45q6fz4ltsntlues3e3gylcu6jsasrwqtape4mvjxf
md1f5przzsafx3dzmj02p3qtffjqpxtu9j5jaevqs7jqje40324vxlw0txswawpxte3zmzspu35hx7lrg2t6
md1f5przz3r3qa03segkpx2s4feevxsd3mta6k0mpf2dw678vdsfa3pu0hswvwhxsh98mjsqpsz8xkj9h16g
md1f5przz3gzl20y277fntzt0qr0j9gl5j8mfdmnm3fvs18a908rvzyr8m4xl7gs9e2t6dyvgv4j

md1f5-przsz-pq3m6-7zzqq-vzrs3-pstuc-w0za5-znwrg-3hcc5-xg5qx-zhs7y-yg2f6-g9kqk-tgkrn-2spp6-tlfzprv6ye
md1f5-przsz-wgyrv-6p2sh-atnt2-a8wzs-2m92f-fsmq-arvqs-qm3lg-xr8tr-r2w5z-jcz3y-jdljk-0qf2g-9tn3u-5jtsz
md1f5-przsz-j7qq7-fkctv-h5jqz-xuepc-cmchn-8u30m-4as45-q6fz4-ltsnt-lues3-e3gyl-cu6js-asrwq-tape4-mvjxf
md1f5-przsz-afx3d-zmj02-p3qtff-fjqp-x-tuj95-jzaev-q57jq-je403-24vxl-fw0tx-swawp-xte3z-mspu3-5hx7l-r2t6
md1f5-prz3-r3qa0-3segk-px2s4-feevx-sd3mt-a6k0m-pf2dw-678vs-dfa3p-u0hsw-nvwhx-sh98m-jsqps-z8xkj-h9l6g
md1f5-prz3-gz1z0-y277f-ntzt0-qr0j9-gl5j8-mfdmnm-3fvs-18a90-8rvzy-r8m4x-l7gs9-e2t6d-yvgv4j

# === Wallet bundle: descriptor, mainnet ===
# Threshold: 2 of 3
# Cosigners:
# @0: (no ms1; watch-only),mk1:c8c3c (73c5da0a @ m/87'/0'/0')
# @1: (no ms1; watch-only),mk1:c8c3d (b8688df1 @ m/87'/0'/0')
# @2: (no ms1; watch-only),mk1:c8c3e (28645006 @ m/87'/0'/0')
# Template: descriptor
# md1: c8c3
# Recovery: any 2 of 3 signing keys + md1 (template card).
note: stdout is watch-only – public keys only, cannot spend
```

3.4 Building from all seeds on one machine (multiple files, one command)

If instead you hold all the seeds yourself, you can build the whole bundle in a single command, reading each seed from its own file with command substitution. This is less safe -- each substituted phrase lands on `argv`, so the toolkit prints a `secret-on-argv` warning for every one -- but it needs no per-device coordination. The three seed files (shown again for reference):

```
$ cat seed0.txt
abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about
```

```
$ cat seed1.txt
legal winner thank year wave sausage worth useful legal winner thank yellow
```

```
$ cat seed2.txt
letter advice cage absurd amount doctor acoustic avoid letter advice cage above
```

Because seeds (not just xpubs) are supplied, this emits the **full secret card set** -- one `ms1` per cosigner -- not the watch-only placeholders of 3.3:

```
$ mnemonic bundle --template wsh-sortedmulti --threshold 2 --network mainnet --slot "@0.phrase=$(cat seed0.txt)" --slot "@1.phrase=$(cat seed1.txt)" --
  ↳ slot "@2.phrase=$(cat seed2.txt)"
warning: secret material on argv (--slot @0.phrase=) - pipe via --slot @0.phrase= to avoid /proc/$PID/cmdline exposure
warning: secret material on argv (--slot @1.phrase=) - pipe via --slot @1.phrase= to avoid /proc/$PID/cmdline exposure
warning: secret material on argv (--slot @2.phrase=) - pipe via --slot @2.phrase= to avoid /proc/$PID/cmdline exposure
# ms1[0] (entropy, BCH-checksummed)
ms10entrsqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqc9sraq34v7f

ms10e ntrsq qqqqq qqqqq qqqqq qqqqq qqcj9 sraq 34v7f

# ms1[1] (entropy, BCH-checksummed)
ms10entrsqplh7lml0alh7lml0alh7lml0als5cclar2zmks6

ms10e ntrsq plh7l ml0al h7lml 0alh7 lml0a ls5cc lar2z mksh6

# ms1[2] (entropy, BCH-checksummed)
ms10entrsqzqgppyqsqgppyqsqzqqlfm7mep84hunu

ms10e ntrsq zqgpy yqszy gpyyq szqpy qyqsq qqlfm 7mep8 4hunu

# mk1[0] (cosigner 0 xpub + origin)
mk1qperpupqqsu3s7denyv8nwverpumn9cdzxlzpcy3zepa59349dcs5n5stvpvk3v8x4eqsdngy2jl4wddt5ac2ptv4fya76tpreapyrdfr
mk1qperpuppnq8kp6xcpqphr7svxwx5ag99s9zfym19v7tcqrexmpdj7jgqgmny8rr0z7vlj7eqzv2486xkkcftrq8

mk1qp erpup qqspu 3s7de nyv8n wverp umnrn chdq5 pcy3z epa59
349dc gs5n5 stvpv k3v8x 4eqsd ngy2j l4wdd t5ac2 ptv4f ya76t
preap yrdfq r
mk1qp erpup pnq8k p6xcp qphr7 svxwk xx5ag 99s9z fym19 v7tcq
rexmp dj7jg qgmny 8rr0z 7vlj7 eqzv2 486xk kcftr zq8

# mk1[1] (cosigner 1 xpub + origin)
mk1qperpupqqsu3s7denyv8nwverpumn9cdzxlzpcy3zepaqxfl630m4as45q6fz4ltsntlues3e3gylcu6jsa6jdz69hy0whcpg2f28j5awu
mk1qperpapp02p3qtffjqpxtu95jzaevqs7jqje40324vxlw0txswawpxte3zmhzp6lprj3ga2lw65h2dd24cuerkpa

mk1qp erpap qqspu 3s7de nyv8n wverp umn9c dzxlz pcy3z epaqx
afl63 0m4as 45q6f z4lts ntlue s3e3g ylcu6 jsa6j dz69h y0whc
pg2f2 8j5aw u
mk1qp erpap p02p3 qtffj qpxtu j95jz aevqs 7jqje 40324 vxlfw
0txsw awpxt e3zmh zp6lr pj3ga 2lw65 h2dd2 4cuer kpa

# mk1[2] (cosigner 2 xpub + origin)
mk1qperp7pqqsu3s7denyv8nwverpumpgv3gqvpcy3zeganecw97sn9g25uukrgxca47at8as54xhd0rkgs57cs78mc8fk6507cs8tnugnnpnf
mk1qperp7ppr4e59ef7u5p038j900ye439hsphes506fra5k3eac5kg0n7jhn3kpzpa6n0lygdlpfeq4a75lc2mrrz7f

mk1qp erp7p qqspu 3s7de nyv8n wverp umnpg v3gqv pcy3z epane
cw97s n9g25 uukrg xca47 at8as 54xhd 0rkgs 57cs7 8mc8f k6507
cs8tn ugnpn f
mk1qp erp7p pr4e5 9ef7u 5p038 j900y e439h sphez 506fr a5k3e
ac5kg 0n7jh n3kpz pna6n 0lygd lpfeq 4a75l c2mrr z7f

# md1 (multisig wallet policy)
md1f5przzspq3m67zzqqvzrs3pstucw0za5znwrg3hcc5xg5qxzhs7y9g2f6g9kqktgkrn2spp6tlfzprv6ye
md1f5przzswgyrv6pz5hatnt2a8wzs2m92ffsrmqarvqsqm3lgxr8trr2w5zjcz3yjd1jk0qf2g9tn3u5jtsz
md1f5przzsj7q7fkctvh5jqzuepcmcchn8u30m4as45q6fz4ltsntlues3e3gylcu6jsasrwatqpe4mvjxf
md1f5przzsafx3dzmj02p3qtffjqpxtu95jzaevqs7jqje40324vxlw0txswawpxte3zmspu35hx7lrg2t6
md1f5przz3r3qa03segkpx2s4feevxsd3mta6k0mpf2dw678vdsd3pu0hswvwhxsh98mjsqpsz8xkh9l6g
md1f5przz3gzl20y277fntz0qr0j9g15j8mfdnm3fvs18a908rvzry8m4xl7gs9e2t6dyvjgv4j

md1f5-przzs-pq3m6-7zzqq-vzrs3-pstuc-w0za5-znwrg-3hcc5-xg5qx-zhs7y-yg2f6-g9kqk-tgkrn-2spp6-tlfzp-rv6ye
md1f5-przzs-wgyrv-6pz5h-atnt2-a8wzs-2m92f-fsrmq-arvqs-qm3lg-xr8tr-r2w5z-jcz3y-jd1jk-0qf2g-9tn3u-5jtsz
md1f5-przzs-j7q7f-fkctv-h5jqz-xuepc-cmchn-8u30m-4as45-q6fz4-ltsnt-lues3-e3gyl-cu6js-asrwq-tqpe4-mvjxf
```



```
mdl1f5-przsz-afx3d-zmj02-p3qtf-fjqpx-tuj95-jzaev-qs7jq-je403-24vxl-fw0tx-swapx-xte3z-mspu3-5hx7l-rg2t6
mdl1f5-przz3-r3qa0-3segk-px2s4-feevx-sd3mt-a6k0m-pf2dw-678vs-dfa3p-u0hsw-nvwhx-sh98m-jsqps-z8xkj-h9l6g
mdl1f5-przz3-gz1z0-y277f-ntzt0-qr0j9-gl5j8-mfdrn-m3fvs-l8a90-8rvzy-r8m4x-l7gs9-e2t6d-yvjgv-4j
```

```
# === Wallet bundle: wsh-sortedmulti, mainnet ===
# Threshold: 2 of 3
# Cosigners:
#   @0: msl:c8c3c,mkl:c8c3c (73c5da0a @ m/87'/0'/0')
#   @1: msl:c8c3d,mkl:c8c3d (b8688df1 @ m/87'/0'/0')
#   @2: msl:c8c3e,mkl:c8c3e (28645006 @ m/87'/0'/0')
# Template: wsh-sortedmulti
# mdl: c8c3
# Recovery: any 2 of 3 signing keys + mdl (template card).
warning: stdout carries private key material (can spend) - redirect or encrypt (e.g. '> file.txt' or '| age -e ...')
```

Only one secret may arrive on stdin, so you cannot replace more than one substitution with the `=` file-redirect form -- a second `=` is rejected:

```
$ mnemonic bundle --template wsh-sortedmulti --threshold 2 --network mainnet --slot @0.phrase=- --slot @1.phrase=- --slot "@2.phrase=$(cat seed2.txt)" <
  ↪ seed0.txt
warning: secret material on argv (--slot @2.phrase=) - pipe via --slot @2.phrase=- to avoid /proc/$PID/cmdline exposure
error: at most one --slot @N.<secret>=- per invocation (single stdin per invocation)
```

4. Card set -> Bitcoin Core wallet descriptor (and how to import)

mnemonic restore --md1 <chunks> reconstructs the watch-only wallet from the **shared md1 card alone** -- no seeds needed. First produce that card from the section-3 wallet (descriptor file shown again) and pull out its md1 chunks:

```
$ cat multisig.desc
wsh(sortedmulti(2,[73c5da0a/87'/0'/0']xpub6DBjFkYsKtK13Q1L35bqddw5z3VGJnf516B3icHrEGEUcCuG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqrXvicM
  ↳ /<0;1>/*,[b8688df1/87'/0'/0']xpub6CbhRzPzY2z7NcCGGjLAJLQ8iRyJufwmdXQs66MxTVUReKb9DpLnVJ5D1qpatZjUuPGTyxf5TYUlvA34YFE9FHB4TvfYmokYLVsyEFZft9
  ↳ /<0;1>/*,[28645006/87'/0'/0']xpub6DB7HNQw6CZoJxN85NuFTPWhi2FagSnexPS1rv3nYQhngkmdHgb7iebYvTfMFKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hjHn2uazz4z
  ↳ /<0;1>/*))
```

```
$ mnemonic bundle --descriptor-file multisig.desc --network mainnet --json | jq -r ".md1[]" > multisig.md1
note: stdout is watch-only - public keys only, cannot spend
```

```
$ cat multisig.md1
md1f5przzspq3m67zzqzvrs3pstucw0za5znwrg3hcc5xg5qxzhs7y9g2f6g9kqktgkgrn2spp6tlfzprv6ye
md1f5przzswgyrv6pz5hatnt2a8wzs2m92ffsrmqarvqsm3lgxr8trr2w5zjcz3yjdLjk0qf2g9tn3u5jtsz
md1f5przzsj7q7fkctvh5jqzxuepcmcn8u30m4as45q6fz4ltsntlues3e3gylcu6j5asrwtqpe4mvjxf
md1f5przzsafx3dzmj02p3qtfjqpuxtj95jzaevqs7jqje40324vxlfw0txswawphte3zmspu35hx7lrg2t6
md1f5przz3rq03se9kpx2s4feevsd3mta6k0mpf2dw678vdsfa3pu0hswvwhxsh98mjsqps28xkjh9l6g
md1f5przz3gzl0y277fntzt0qr0j9gl5j8mfdnm3fvs18a908rvzyr8m4xl7gs9e2t6dyvjgv4j
```

Restore reconstructs the wallet from exactly those chunks. The default form prints the descriptor and first address (note the address matches section 3 -- same wallet -- while the descriptor *string* differs because the md1 card stores each key as a depth-0 master xpub, an equivalent serialisation):

```
$ mnemonic restore --network mainnet $(sed "s/~/--md1 /" multisig.md1)
2-of-3 multisig restore
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: wsh(sortedmulti(2,[73c5da0a/87'/0'/0']
  ↳ xpub661MyMwAQBcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzw8TBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPWDUSNJsY5JsgVBQoD6KrbdCLL/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub661MyMwAQBcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXyukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmrEkfhjzghCUEcvJBhXS5bk4RW/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub661MyMwAQBcEddy4jr5EtEhQBctfscE6a99DGLr2cW4HnnmBsXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/<0;1>/*))#yjp7hj7w
first recv: bc1qkssenl2m6t3aynza394sr9m86vt6md2v76kj52jun2x1wrdeaa4q84qtpl
cosigner @0: 73c5da0a [87'/0'/0'] from md1 (not independently verified)
cosigner @1: b8688df1 [87'/0'/0'] from md1 (not independently verified)
cosigner @2: 28645006 [87'/0'/0'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only - public keys only, cannot spend
```

Add --format bitcoin-core for a ready-to-import importdescriptors request array (external .../0/* + change .../1/*):

```
$ mnemonic restore --network mainnet $(sed "s/~/--md1 /" multisig.md1) --format bitcoin-core
[
  {
    "active": true,
    "desc": "wsh(sortedmulti(2,[73c5da0a/87'/0'/0']
      ↳ xpub661MyMwAQBcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzw8TBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPWDUSNJsY5JsgVBQoD6KrbdCLL/0/*,[b8688df1/87'/0'/0']
      ↳ xpub661MyMwAQBcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXyukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmrEkfhjzghCUEcvJBhXS5bk4RW/0/*,[28645006/87'/0'/0']
      ↳ xpub661MyMwAQBcEddy4jr5EtEhQBctfscE6a99DGLr2cW4HnnmBsXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/0/*))#y65a0dtg",
    "internal": false,
    "range": [
      0,
      999
    ],
    "timestamp": 0
  },
  {
    "active": true,
    "desc": "wsh(sortedmulti(2,[73c5da0a/87'/0'/0']
      ↳ xpub661MyMwAQBcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzw8TBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPWDUSNJsY5JsgVBQoD6KrbdCLL/1/*,[b8688df1/87'/0'/0']
      ↳ xpub661MyMwAQBcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXyukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmrEkfhjzghCUEcvJBhXS5bk4RW/1/*,[28645006/87'/0'/0']
      ↳ xpub661MyMwAQBcEddy4jr5EtEhQBctfscE6a99DGLr2cW4HnnmBsXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/1/*))#k0gfvz2t",
    "internal": true,
    "range": [
      0,
      999
    ],
    "timestamp": 0
  }
]
2-of-3 multisig restore
CONFIRM: verify each cosigner fingerprint against your records before importing the payload above.
descriptor: wsh(sortedmulti(2,[73c5da0a/87'/0'/0']
  ↳ xpub661MyMwAQBcFrooZ2966EcDmVX5MoFXZhuJqXTudvJzw8TBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPWDUSNJsY5JsgVBQoD6KrbdCLL/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub661MyMwAQBcEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXyukNGMK2XFRbAaCLYEJDUUfNoBxTNa68i5MYhqmrEkfhjzghCUEcvJBhXS5bk4RW/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub661MyMwAQBcEddy4jr5EtEhQBctfscE6a99DGLr2cW4HnnmBsXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/<0;1>/*))#yjp7hj7w
first recv: bc1qkssenl2m6t3aynza394sr9m86vt6md2v76kj52jun2x1wrdeaa4q84qtpl
```

```
cosigner @0: 73c5da0a [87'/0'/0'] from md1 (not independently verified)
cosigner @1: b8688df1 [87'/0'/0'] from md1 (not independently verified)
cosigner @2: 28645006 [87'/0'/0'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only – public keys only, cannot spend
```

Import into Bitcoin Core: save the array, create a blank descriptor wallet, and load it (these run against your own node, so their output is not shown here):

```
$ mnemonic restore --network mainnet $(sed "s/^/--md1 /" multisig.md1) --format bitcoin-core > wallet.json
$ bitcoin-cli -named createwallet wallet_name="multisig-2of3" disable_private_keys=true blank=true descriptors=true
$ bitcoin-cli -rpcwallet="multisig-2of3" importdescriptors "$(cat wallet.json)"
$ bitcoin-cli -rpcwallet="multisig-2of3" getnewaddress
```

Tips: `--timestamp now` skips the rescan for a fresh wallet (default 0 rescans from genesis); `--range 0,4999` widens the gap limit; `--bitcoin-core-version 24` targets older Core. `restore` also emits `--format descriptor` (the bare `wsh(...)#checksum`) for other wallets.

5. Custom degrading-miniscript wallet (distinct keys per tier) + watch-only export

A four-tier vault. **Each tier uses its own distinct key set (no key reuse)**, deliberately mixing all four Bitcoin timelock kinds:

Tier	Spend condition	Timelock kind
1	3-of-3 (K0,K1,K2) + secret word	absolute height -- after(1000000)
2	2-of-3 (K3,K4,K5) + secret word	absolute time -- after(1893456000)
3	both K6 and K7	relative blocks -- older(65535)
4	any 1 of K8,K9,K10	relative time -- older(4255898)

That is **11 distinct keys** (3+3+2+3). Absolute locks (*after*) count from the chain's height/clock; relative locks (*older*) count from each coin's own confirmation. Encodings:

- after(1000000) -- absolute **block height** 1,000,000 (BIP-65; values below 500,000,000 are heights).
- after(1893456000) -- absolute **Unix time** = 2030-01-01 00:00 UTC (values at/above 500,000,000 are timestamps).
- older(65535) -- relative **blocks**: 65,535 blocks (~455 days). This is the largest safe relative-block lock; older(65536) would be BIP-68 consensus-masked to zero, and the toolkit warns if it sees one.
- older(4255898) -- relative **time**: BIP-68's time flag (bit 22) set, plus 61,594 units x 512 s ~ 365 days.

5.1 The secret word (a hashlock shared by tiers 1 and 2)

Reusing a *hash* across tiers is fine -- it is not a key. The secret word is `opensessame`; the descriptor commits to $H = \text{sha256}(\hookrightarrow \text{sha256}(\text{word}))$ and a spend reveals the 32-byte preimage $X = \text{sha256}(\text{word})$:

```
$ python3 -c "import hashlib; w='opensessame'; X=hashlib.sha256(w).digest(); print('preimage X =', X.hex()); print('hash H      =', hashlib.sha256(X).hexdigest())"
preimage X = cd6f70a4440de96063f8dbd7a4a3bbcff6b993af6c83e08d87b359e0760ca9c6
hash H      = a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad
```

5.2 The guided builder caps complexity -- use the raw descriptor path

`mnemonic build-descriptor` runs a satisfiability + cost preview that it **bounds** for funds-safety. An 11-key, 4-branch policy exceeds that envelope, so the guided builder refuses and points you at the raw `--descriptor` path. The policy-tree spec it reads:

```
$ cat policy.json
{
  "schema_version": 1,
  "wrapper": "wsh",
  "root": {
    "or_i": [
      {
        "and_v": [
          {
            "wrap": {
              "w": "v",
              "sub": {
                "after": 1000000
              }
            }
          },
          {
            "and_v": [
              {
                "wrap": {
                  "w": "v",
                  "sub": {
                    "sha256": "a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad"
                  }
                }
              },
              {
                "multi": {
                  "k": 3,
                  "keys": [
                    "[73c5da0a/84'/0'/0']xpub6CatWdiZiodmUeTdp8LT5or8nmbKNcuYvz7WYksVFkKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLoP2fMFMKHPJ4ZeZXYVUhlV1VMrjPC7PW6V"
                  ]
                }
              }
            ]
          }
        ]
      }
    ]
  }
}
```

```

    " [73c5da0a/84'/0'/1']xpub6CatWdiZiodmYVtWLTtEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEZv3quZSJKA5dh5Lo2PbubMGxCfZtRthV6ST2qqUL9w3HSzcUn
    ↪ ",
    " [73c5da0a/84'/0'/2']xpub6CatWdiZiodmbNGqcQxxjGN165QxTU4PwNNi9WrijYgYf7VxcmuFxsRw3foczLgRDbjDjJbqZhPCTfkcaWmL9BuSw98ybKKJtcHgWeryy6"
  ]
}
}
]
}
},
{
  "or_i": [
    {
      "and_v": [
        {
          "wrap": {
            "w": "v",
            "sub": {
              "after": 1893456000
            }
          }
        },
        {
          "and_v": [
            {
              "wrap": {
                "w": "v",
                "sub": {
                  "sha256": "a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad"
                }
              }
            },
            {
              "multi": {
                "k": 2,
                "keys": [
                  " [73c5da0a/84'/0'/3']
                  ↪ xpub6CatWdiZiodmdHurRokjbycCrxddTDJgTsyEAaQfKjKwbwUi79LAWG5gHjMCQB7BeJc47MkubXuZdf45JZHK1qcr1GZ5EwREUDVdLVdPKEC",
                  " [b8688df1/84'/0'/0']
                  ↪ xpub6DNfJehqF1LLUs9kwaqDu12Ajpz9psYVtbGhTykQo1CYdkkqV2vAyR4DiwXSTTDuJwHzVy1AtV6ENGKwgbLWqa4wXMZR4ZmdpRjQB65EgTV",
                  " [b8688df1/84'/0'/1']
                  ↪ xpub6DNfJehqF1LLUsoh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvqqFDLWuyCivJ4eMDxgFtiTxnazgp8rYVzQSKR3L8EGRs9asBarqpe"
                ]
              }
            }
          ]
        }
      ]
    },
    {
      "or_i": [
        {
          "and_v": [
            {
              "wrap": {
                "w": "v",
                "sub": {
                  "older": 65535
                }
              }
            },
            {
              "multi": {
                "k": 2,
                "keys": [
                  " [b8688df1/84'/0'/2']
                  ↪ xpub6DNfJehqF1LLUwZ1DrFcQN365CDMPLSsrqpSwaz4eRrtYz9qWdsVH9JqsHpQ6yNGdG4VXGKbLgcxKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51",
                  " [b8688df1/84'/0'/3']
                  ↪ xpub6DNfJehqF1LLUxb8gRSstbR9LcAxWgwD4V678z5FZ7BLftWzwQCt3yZb5eW4U8AVJwbKcZ2ieggjvKv2xggJpMBRkk3CE7bz7g4uMV7Qp3TU"
                ]
              }
            }
          ]
        },
        {
          "and_v": [
            {
              "wrap": {
                "w": "v",
                "sub": {
                  "older": 4255898
                }
              }
            },
            {
              "multi": {

```

Running the guided builder on it:

(For a policy *within* the envelope -- fewer keys -- `build-descriptor --spec` validates and emits it for you.) For arbitrarily complex policies you hand the miniscript descriptor straight to `export-wallet / bundle`. The hand-written descriptor file:

Validate and canonicalise it (this adds the BIP-380 checksum). The full canonical descriptor, with every xpub in full

First receive address (Mainnet), via the BSMS record:

14

```

↳ (65535),multi(2,[b8688df1/84'/0'/2']
↳ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqshpQ6yNGdG4VXGKbLgCcxKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
↳ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzwQCT3yZb5eW4U8AVJwbKcZ2ieggjvKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*),and_v(v:older(4255898)
↳ ,multi(1,[28645006/84'/0'/0']xpub6DBBzvudcQg2nS4tHskrm7F6kXL6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpU8p2Xx4wWQK7iGBycAm
↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvudcQg2sPDRWpqfEn5VzPiwrD1zNes4aHmNUog9Jmc2fc2J5fM2E39YzY2iakmpWRpa3rXhzGNmd4GKiJEZxmTcftQpWBGd9ihVks
↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABfJ4j8TRxUnRd5eGgEXfgPj63xcuTV9uny7pQ8Fbu3XKcN9rNxcNGRaDT9BD1gkBGZ
↳ /<0;1>/*)))#4ld0crxa
/0/*,/1/*
bc1q4g7564xxd9hj68hquw5e558cqafhsklerkr0asfzqp6uq74veesrp6qss
note: stdout is watch-only – public keys only, cannot spend

```

5.3 Engrave the card set

Because every key is **distinct**, this is a valid BIP-388 wallet policy, so -- unlike a key-reusing policy -- `bundle` will engrave it. With only public xpubs supplied, the result is watch-only (the `ms1` cards are empty placeholders). The descriptor file it reads:

```

$ cat policy.desc
wsh(or_i(and_v(after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,[73c5da0a/84'/0'/0']
↳ xpub6CatWdiZiodmUeTp8LT5or8nmBKNcuyvz7WysVFkKB4RHCD3YuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUhLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
↳ xpub6CatWdiZiodmYVtWLTtEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEZv3quZSJKASdh5Lo2PpubMGxCfZtRthV6ST2qquL9w3HSzcUn/<0;1>/*,[73c5da0a/84'/0'/2']
↳ xpub6CatWdiZiodmbNgqcQxxjGN165QxTU4PwNNi9WrijYgyf7VxcmuFosRw3focZLgRDbjDjJbqZhpCTfKcaWmL9Bu5w98ybKJtchGweryy6/<0;1>/*)),or_i(and_v(after
↳ (1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(2,[73c5da0a/84'/0'/3']
↳ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsyEaaQfKjKwBwUi79LAWG5gHjMCQB7BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLdVdPKEC/<0;1>/*,[b8688df1/84'/0'/0']
↳ xpub6DNfJehqF1LU59kwaqDu12Ajz9psYVtbGhTykQo1CYdkkqV2yAr4DiWxSTTDujWHzVy1AtV6ENGKwGwLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
↳ xpub6DNfJehqF1LUsoh1a4XLG3yuFY1oCRJd6Bkba97xvW65wvY71Nc6LvqfDLWuyCiv4eMDxgFtiTnxazgp8rYVzQSKR3L8EGRs9asBarqpe/<0;1>/*)),or_i(and_v(older
↳ (65535),multi(2,[b8688df1/84'/0'/2']
↳ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqshpQ6yNGdG4VXGKbLgCcxKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
↳ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzwQCT3yZb5eW4U8AVJwbKcZ2ieggjvKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*),and_v(v:older(4255898)
↳ ,multi(1,[28645006/84'/0'/0']xpub6DBBzvudcQg2nS4tHskrm7F6kXL6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpU8p2Xx4wWQK7iGBycAm
↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvudcQg2sPDRWpqfEn5VzPiwrD1zNes4aHmNUog9Jmc2fc2J5fM2E39YzY2iakmpWRpa3rXhzGNmd4GKiJEZxmTcftQpWBGd9ihVks
↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABfJ4j8TRxUnRd5eGgEXfgPj63xcuTV9uny7pQ8Fbu3XKcN9rNxcNGRaDT9BD1gkBGZ
↳ /<0;1>/*)))#4ld0crxa

```

The watch-only card set:

```

$ mnemonic bundle --descriptor-file policy.desc --network mainnet
# ms1 (omitted – descriptor watch-only mode)

# mk1[0] (cosigner 0 xpub + origin)
mk1qptfcrzqq945upmkp8qwastrfcrhvh6wqamqkns8wc95upmkp8qwastrfcrhvh6wqamqkns8wc95upmkpeutks2qzvzgs7hd8pnnmcv56c4u
mk1qptfcrzpgk08aueetmd998g9tyxuae9vxn38f9gtrp98q8s808l6szjxkt6r83rk2jg0bcqns0f30mtxzd65mwvcur9usv8qhcmu69gql0zrc
mk1qptfcrz74hwqxdp883jehp5tdrfaf0n5zdfhw6425sqe8pgsyxfjtjtf4

mk1qp tfczr qqs94 5upmk p8qwa astfc rhvz6 wqamq kns8w c95up
mkp8d qwastrfcrhvh z6wqa mqkns 8wc95 upmkp eutks 2qvzgs s7hd8
pnnmc v56c4 u
mk1qp tfczr pkg08 auetm d998g 9tyxu ae9vx n38f9 gtrp9 8qk8s
08l6s zjxkj t6r83 rk2jg 0bcns 0f30m txzd6 5mwvcur9us v8qhc
mu69g ql0zr c
mk1qp tfczr z74hw qxqdp 083je hp5td rfa0n 5zdfh w6425 sqe8p
gsyxf jtjtf4

# mk1[1] (cosigner 1 xpub + origin)
mk1qptfcrzqq945upmkp8qwastrfcrhvh6wqamqkns8wc95upmkp8qwastrfcrhvh6wqamqkns8wc95upmkpeutks2lcpag7xw0uuvz59ukc0
mk1qptfcrzpszgqgzyqsqqgzyqsqqgzy3zepulhn90du6setz7wccgw0mnt0jarage0ptr0vxf3c9895ghkha0gk58rcwcdclrvxvprnta
mk1qptfcrz74hwqxdp883jehp5tdrfaf0n5zdfhw6425sqe8pgsyxfjtjtf4

mk1qp tfczr qqs94 5upmk p8qwa astfc rhvz6 wqamq kns8w c95up
mkp8d qwastrfcrhvh z6wqa mqkns 8wc95 upmkp eutks 2lcpa g7xw
00uvz 59ukc 0
mk1qp tfczr pszgq qzyqs qqgzy ypsqz qqzqy 3zepu lhn90 du6se
tz7wc ggw0m nt0ja rage0 ptr0v xf3c9 895gh kha0g k58rc wcdc
lrvxv vprnt a
mk1qp tfczr zdp8 2uxnp rxcue jaent sxshg r9vud frx5c c7npa
egptf 64nsu d0nnn t334x du0p8 3dn95 g54tz

# mk1[2] (cosigner 2 xpub + origin)
mk1qptfcrzqq945upmkp8qwastrfcrhvh6wqamqkns8wc95upmkp8qwastrfcrhvh6wqamqkns8wc95upmkpeutks2lcpag07cz8k5qayd8z5
mk1qptfcrzpszgqgzyqsqqgzyqsqqgzy3zepulhn90d76enfqwtug4zdpn8r94nrhgnwyc6l2errqlch68l94cdsr0vxxvngvzwq0fhe86j
mk1qptfcrz74hwqxdp883jehp5tdrfaf0n5zdfhw6425sqe8pgsyxfjtjtf4

mk1qp tfczr qqs94 5upmk p8qwa astfc rhvz6 wqamq kns8w c95up
mkp8d qwastrfcrhvh z6wqa mqkns 8wc95 upmkp eutks 2lcpa g07cz
z8k5q ayd8z 5
mk1qp tfczr pszgq qzyqs qqgzy ypsqz qqzqy 3zepu lhn90 d76en
fqwtu ug4zd pn8r9 4nrhg nwycc 6l2er rqlch 68l94 cdsr0 vxvng
vzwq0 fhe86 j
mk1qp tfczr zqup q34fl c5lnh z7dzk jmwzf htn2t f9tp8 385pw
pa2rh 2p9a6 flzxm hl43h hx3cm gcjp6 25l8l

```

```
# mk1[3] (cosigner 3 xpub + origin)
mk1qptfcqzqs945upmkpd8qwestfcrhvv6wqamqkns8wc95upmkpd8qwestfcrhvv6wqamqkns8wc95upmkpeutks2lcpagqk6f7csue6uzgna
mk1qptfcqzpsqzqgqzyqsqgqzyrszqgqzy3zepulhn90d6awkdsm10jm7xjylhyyp7esaklwkwkasq24gnq3578mne3sjr64gdauj68uhy768d
mk1qptfcqzqzdp0dmvslcy68sh0lw93nwwxth488eqjvn67gqjpx7fyf84xj7zjfc29xjtagpghsxz5xlu0rx

mk1qp tfcqz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up
mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkp eutks 2lcpa gqk6f
7csue 6uzgn a
mk1qp tfcqz pszqg qzyqs zqgqz yrszq gqzqy 3zepu lhn90 d6awk
dsm10 jm7xj ylhy p7esa klwk kasq2 4gnq3 578mn e3sjr 64gda
uj68u hy768 d
mk1qp tfcqz zqp0 dmvs1 cy68s h0lw9 3nwwx th488 eqjvn 67gqj
px7fy f84xj 7zjfc 29xjt agpg h sxz5x lu0rx

# mk1[4] (cosigner 4 xpub + origin)
mk1qptfc8zqs945upmkpd8qwestfcrhvv6wqamqkns8wc95upmkpd8qwestfcrhvv6wqamqkns8wc95upmkzux3r03qvzgsnwq0e2f7mmwlennx
mk1qptfc8zpkg0w55t7uh3mhpjgm2ctzuevmwdfap2f5z19lkdftfh372tg8y9k3d35qm006q30450k4vmqvakpywv7pkdajnnrd36ymdaqksnv
mk1qptfc8z2m36y9wa86gpqpq5ypppy22y3ww23fxkwgjcskxqsga8sgz6t1

mk1qp tfc8z qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up
mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkz ux3r0 3qvzgsnwq0
e2f7m mwlnn x
mk1qp tfc8z pkg0w 55t7u h3mhp gjm2c tzuev mwdfp ap2f5 z19lk
dtfh3 72tg8 y9k3d 35qm0 06q30 450k4 vmqva kpywv 7pkda jnnrd
36ymd aqksn v
mk1qp tfc8z zm36y 9wa86 gpqpq 5yppp y22y3 ww23f xkwgjc skxq
sga8s gz6t1

# mk1[5] (cosigner 5 xpub + origin)
mk1qptfczqs945upmkpd8qwestfcrhvv6wqamqkns8wc95upmkpd8qwestfcrhvv6wqamqkns8wc95upmkzux3r03lcpagnlnxhkxfprlea74
mk1qptfczpszqgqzyqsqgqzyrszqgqzy3zepa6j30mj2615afq22n2ct2h3zwrjpsfeyfqr5zrzs70jdmqtqv7wwt026a3fmdmqr0khu23
mk1qptfczqzdp6ttmcl57ac6vels0u3jj86dd5pe7m355rttw8r3a2s45tszcmrx9xa5qyec4g3crut8d54

mk1qp tfcz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up
mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkz ux3r0 3lcpa gnlnx
hkxfp rlea7 4
mk1qp tfcz pszqg qzyqs zqgqz ypsqz gqzqy 3zepa 6j30m jj261
5afq2 2n2ct 2h3zw rjpse fyfqr 5zrzs 70jdm tqvx7 wwt0z 6a3fm
fdmqr 0khu2 3
mk1qp tfcz zdp6 ttmc1 y57ac 6vels 0u3jj 86dd5 pe7m3 55rtt
w8r3a 2s45t szcmr x9xa5 qyec4 g3cru t8d54

# mk1[6] (cosigner 6 xpub + origin)
mk1qptfc9zqs945upmkpd8qwestfcrhvv6wqamqkns8wc95upmkpd8qwestfcrhvv6wqamqkns8wc95upmkzux3r03lcpagz84k67v3r27gvvyw
mk1qptfc9zpszqgqzyqsqgqzyrszqgqzy3zepa6j30mjemst4u6j4atq6uhd0m3m5l9lvu9cde6pcswdr73ejs4lts9qugnu fy3yg6r2y0f
mk1qptfc9zvgpqdy4ej7rexgzc7vsch5dxs9e587e8u22q38ujtrt541m26dq30kfqvm7wyvxtmzy38suff

mk1qp tfc9z qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up
mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkz ux3r0 3lcpa gz84k
67v3r 27gvvy w
mk1qp tfc9z pszqg qzyqs zqgqz ypsqz gqzqy 3zepa 6j30m jemst
4u6j4 atq6u hd0m3 m5l9l vu9cd e6pcs wdr7 3ejs 4lts9 qugnu
fy3yg 6r2y0 f
mk1qp tfc9z zvgpq dye4j 7rexg zc7vs ch5dx s9e58 7e8u2 2q38u
jtrt5 4lm26 dq30k fqvm 7wyvxtmzy3 8suff

# mk1[7] (cosigner 7 xpub + origin)
mk1qptfcyqs945upmkpd8qwestfcrhvv6wqamqkns8wc95upmkpd8qwestfcrhvv6wqamqkns8wc95upmkzux3r03lcpagd0hpxp2e6dx8r48
mk1qptfcyqzpsqgqzyqsqgqzyrszqgqzy3zepa6j30mjstc3wnact9lauqxytd7f9xhqmtmlma4zhsvtx5p6e3s3rggp2ce7mxam45j76f5qd
mk1qptfcyqz3qpzlzuww4vzvcksequdfl7d8u34vrr7t6kyjwlagkep2ywk6g2q5h3z3tv9m06y2yd6kfgs

mk1qp tfcy qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up
mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkz ux3r0 3lcpa gd0hp
xp2e6 dx8r4 8
mk1qp tfcy pszqg qzyqs zqgqz yrszq gqzqy 3zepa 6j30m jstc3
wnact 9lauq xytd7 f9xhq mtmlm a4zhs vtx5p 6e3s3 rggp2 ce7mx
am45j 76f5q d
mk1qp tfcy z3qpz lzuww w4vzv ckseq udf17 d8u34 vrr7t 6kyjw
lagke p2ywk 6g2q5 h3z3t vy9m0 6y2yd 6kfgs

# mk1[8] (cosigner 8 xpub + origin)
mk1qptfctzqs945upmkpd8qwestfcrhvv6wqamqkns8wc95upmkpd8qwestfcrhvv6wqamqkns8wc95upmkq5xg5qxqvzgs9n3l96x69pda2ys
mk1qptfctzpkg0dqcwzpsyrk9a808d5f0gennj3x9vjds63rqgzndx78e533rulqyzd58cmvq4nudnky9jhqsh5zpemaskg7pdgxcftukrgv6n5
mk1qptfctz2t4c4w6rtnmhnt889lthvne3ktxs5encnny6spl8sgsf53tn98

mk1qp tftz qqs94 5upmk pd8qw astfc rhvz6 wqamq kns8w c95up
mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkq 5xg5q xqvzgs9n3l
96x69 pda2y s
mk1qp tftz pkg0d qcwzp syrk9 a808d 5f0ge nnj3x 9vjds 63rqg
zndx7 8e533 rulqy zd58c mvq4n udnky 9jhqs h5zpe maskg 7pdgx
cftuk rgv6n 5
mk1qp tftz zt4c4 w6rtm nhnt 889lthvne3 kt5e ncny 6spl8
gsqf5 3tn98

# mk1[9] (cosigner 9 xpub + origin)
mk1qptfc2zqs945upmkpd8qwestfcrhvv6wqamqkns8wc95upmkpd8qwestfcrhvv6wqamqkns8wc95upmkq5xg5xlcpgagz2ktxfdleumyfr
```



```
mk1qptf2zpszqqgzyqszqgqzypszqgqzy3zepa5rpgcx0jlag0vc85ztty3las62hfw73f3j8zx6t3lmtpe8skfujv0xfu3hslfcdrrrp8me0
mk1qptfc2zzlcpn80c5zmmvwxqknngp3d279fuywf7zfw5kqe8wv93v9fvn0m39u65adlnvgwullstspxm44n

mk1qp tfc2z qqs94 Supmk pd8qw astfc rhvz6 wqamq kns8w c95up
mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkq 5xg5q xlcpa g9zzk
txfdl eumyf r
mk1qp tfc2z pszqg qzyqs zqgqz ypsqz gqzqy 3zepa 5rpgc x0jla
g0vc8 5ztty 3las6 2hfw7 3f3j8 zx6t3 lmtpe 8skfu jv0xf u3hsl
fcdrr rp8me 0
mk1qp tfc2z zlcpn 80c5z mmvwx qknng p3d27 9fuyw f7zfw 5kqe8
wv93v 9fvn0 m39u6 5adln vgwul lstsp xm44n

# mk1[10] (cosigner 10 xpub + origin)
mk1qptfcfzqqs945upmkpd8qwastfcrhvz6wqamqkns8wc95upmkpd8qwastfcrhvz6wqamqkns8wc95upmkq5xg5qlcpag56yxxwr4asa24nc
mk1qptfcfzpszqqgzyqszqgqzypszqgqzy3zepa5rpgcxwd86v5gunengsxd2x3nhwsl8dfjjs79jwhy7qnx8r24vls4gl05j6uxcc7dl2rjmec
mk1qptfcfzzypsarcmr17tgutcg8uf4m7g fcd8l fv33knprwthl06am5gdw9a9nfx4jz8l2kjr0p0ur43e8j

mk1qp tfcfz qqs94 Supmk pd8qw astfc rhvz6 wqamq kns8w c95up
mkpd8 qwast fcrhv z6wqa mqkns 8wc95 upmkq 5xg5q xlcpa g56yx
xwr4a sa24n c
mk1qp tfcfz pszqg qzyqs zqgqz ypsqz gqzqy 3zepa 5rpgc xwd86
v5gun engsx d2x3n hws18 dfjjs 7jwhy 7qnx8 r24vl s4gl0 5j6ux
cc7dl 2rjme c
mk1qp tfcfz z2yps arcmr 17gut tcg8u f4m7g fcd8l fv33k nprwt
hl06a m5gdw 9a9nf x4jz8 l2kjr 0p0ur 43e8j

# md1 (multisig wallet policy)
md1ffumxts9z3m6jzqaafrp802gfgaafp9nh4yypm6jzxxw75sj3m6jzt802ggrh4yypvaafp9genj3xrxjhltxq
md1ffumxtsqgpxp2v6mq8q5ysvz6a4pxuusyhzunu8xqz8naa2r2akwukvgm42gws7xkx9k7ctgmfdfe9crr4q
md1ffumxts5smngy26xzqyn9xddhpk7cspdxw6snnwgtz4wf7rnnqpre774p4wm8wtxyd64yxex7jxulxvqvk
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md1ffumxtp38xdwq6zaqv4n34yv6nrr6v8h9q9d82kwr347wwd9mtxdypee03z5f5xvuvkkvwu4eplqpedw7ye5
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md1ffumxtp3nqms5fd4v93wvkdhx557s4y6p0jlmx45mcl995rjzmgkc6qdhhhaqgh668m2kdsxy5ru7p2fk5qgs
md1ffumxtp3nfmvzgueurvm8m8gs4m5lffqyqzssssy3fjg9ee29zj26l5afq22n2ct2h3zwrj46trnchtrjptz
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md1ff-umxtp3-9z3m6-jzqaa-fpr80-2gfga-afp9n-h4yyp-m6jzx-w75sj-3m6jz-t802g-grh4y-yvaaf-p9gen-j3xrxj-hltjx-q
md1ff-umxtp3-gqpxp-zv6mq-8q5ys-zv6a4-pxuus-yh2un-u8xqz-8naa2-r2akw-ukvgm-42gws-7xkx9-k7ctq-mfdfe-9crr4-q
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md1ff-umxtp3-euccf-paqzp-0dmvs-lcy68-sh0lv-93nnw-xth48-8eqjv-n67gq-jpx7f-yf84x-j7zjf-c2wr6-04yur-u2eq3-4
md1ff-umxtp3-nqms5-fd4v9-3wvkd-hx557-s4y6p-0jlmx-45mcl-995rj-zmgkc-6qdhh-aqgh6-68m2k-dsxy5-ru7p2-fk5qg-s
md1ff-umxtp3-fmvzg-ueurvm-m8m8-gs4m5-lffqyq-zssss-y3fg-j9ee2-9zj26-l5afq-22n2c-t2h3z-wrj46-trnch-trjpt-z
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md1ff-umxtp3-f8skf-ujv0x-falsr-xwl3g-9hkcc-vpd8x-srz64-u2ncg-unuyj-afvpj-wuctz-c2jex-lhznk-tnnzz-u3sml-7
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md1ff-umxtp3-4l-luk3-chss0-cncthusns607jerrdxzxuh07l4mhgs6ut6pxjq3gs642g7-8tg45

# === Wallet bundle: descriptor, mainnet ===
# Threshold: 3 of 11
# Cosigners:
# @0: (no ms1; watch-only),mk1:5a703 (73c5da0a @ m/84'/0'/0')
# @1: (no ms1; watch-only),mk1:5a702 (73c5da0a @ m/84'/0'/1')
# @2: (no ms1; watch-only),mk1:5a701 (73c5da0a @ m/84'/0'/2')
# @3: (no ms1; watch-only),mk1:5a700 (73c5da0a @ m/84'/0'/3')
# @4: (no ms1; watch-only),mk1:5a707 (b8688df1 @ m/84'/0'/0')
# @5: (no ms1; watch-only),mk1:5a706 (b8688df1 @ m/84'/0'/1')
```


6. Taproot version of the degrading wallet

wsh(...) reveals the whole policy on every spend. **Taproot** gives a cooperative **key-path** spend (cheap, private, looks like single-sig) and splits the fallbacks into **script-tree leaves**, so a spend reveals only the leaf it uses. We keep the same four timelock/hash/multisig tiers as fallbacks and add a distinct cooperative internal key k_{int} ([.../84'/0'/4/1]). Taproot multisig uses `multi_a`, not `multi`. **12 distinct keys** in total.

6.1 Why depth-1 (the one-tier-per-leaf limit on master)

The tidiest layout is one tier per leaf (4 leaves), but that is a **depth-2** taptree, and the shipped rust-miniscript pin mis-formats depth->=2 taptrees (the PR-#953 bug). The toolkit refuses such a descriptor up front rather than emit a malformed one. The four-leaf (depth-2) descriptor file:

```
$ cat taproot-4leaf.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNTnqr2UHCBEsCoL3eEFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{and_v(v:
  ↳ after(1000000),and_v(v:sha256(a84dce40975727c398023c fbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↳ xpub6CatWdiZiodmUeTDp8LT5or8nmbKNcuyvz7WYksVfKKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLOp2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↳ xpub6CatWdiZiodmYVtWLTtEQsAg1H9oo5lBmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZS3KA5dh5Lo2PbubMGxCfZtRthV6S2tqquL9w3HSzcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↳ xpub6CatWdiZiodmNGqCQxxjGN165QxTU4PwNni9WriJYgYf7VxcmuFosRw3focZLgRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgwWeryy6/<0;1>/*)),and_v(v:after
  ↳ (1893456000),and_v(v:sha256(a84dce40975727c398023c fbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↳ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsYEAaQfKjKwBuU179LAWG5GhJMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLVdPkEC/<0;1>/*,[b8688df1/84'/0'/0']
  ↳ xpub6DNfJehqF1LU9kwaqDu12Ajpz9psYVtbGhTykQo1CYdkkqV2vAyR4D1wXSTTDujWhZvY1AtV6ENGKwGwBLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↳ xpub6DNfJehqF1LUsoh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvKqFDLWuyCivJ4eMDxgFtiTxnazgp8rYvZQSKR3L8EGRS9asBarqpe/<0;1>/*)),{and_v(v:older
  ↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↳ xpub6DNfJehqF1LUwZ1DrFCQ365CDMPLESsrrqpSwaz4eRrtYz9qWdsVH9JqshPQ6yNgdG4VXGKbLgCkKkXtWtB5B4iCuHcmjomqR8z6NNXnpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↳ xpub6DNfJehqF1LUxb8gRSstbR9LcAXwGwD4V678z5FZ7BLftWzWQct3yZb5eW4U8AVjwbKcZ2iegjvKvX2gxpJmBRkk3CE7bz7g4uMV7Qp3TU/<0;1>/*)),and_v(v:older(4255898)
  ↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBbzvudcQg2nS4tH5krm7FGkXl6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpu8p2Xx4wWQK7iGBycAm
  ↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBbzvudcQg2sPDRWpqfEn5VzPiwrd1zNes4aHmnnUog9Jmc2fc2J5fM2E39Yz2iakmpWRpa3rXhZGnm4dGKiJZEXmTCftQpWBGd9ihVks
  ↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBbzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABF4j48TRxUnRd5eGgEXfgP63xcuTV9uny7pQBFbu3KKcN9rNxcNGRADt9BD1gkBGZ
  ↳ /<0;1>/*)))})
```

Asking the toolkit to export it:

```
$ mnemonic export-wallet --descriptor "$(cat taproot-4leaf.desc)" --format descriptor --network mainnet
error: export-wallet script-type derive: taptree branch must have 2 children, but found 1
```

So we use a **depth-1** tree (2 leaves) and pack two tiers per leaf with `or_i`: Leaf A = tier 1 or tier 2 (the absolute-timelock + secret-word tiers); Leaf B = tier 3 or tier 4 (the relative-timelock tiers). (A rust-miniscript release > 13.1.0 containing #953 reopens deep trees -- tracked in FOLLOWUP [taproot-coverage-cycle-on-miniscript-gt-13-1-0.](#))

6.2 Build + validate

The hand-written depth-1 `tr(...)` descriptor file:

```
$ cat taproot.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNTnqr2UHCBEsCoL3eEFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{or_i(
  ↳ and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023c fbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↳ xpub6CatWdiZiodmUeTDp8LT5or8nmbKNcuyvz7WYksVfKKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLOp2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↳ xpub6CatWdiZiodmYVtWLTtEQsAg1H9oo5lBmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZS3KA5dh5Lo2PbubMGxCfZtRthV6S2tqquL9w3HSzcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↳ xpub6CatWdiZiodmNGqCQxxjGN165QxTU4PwNni9WriJYgYf7VxcmuFosRw3focZLgRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgwWeryy6/<0;1>/*)),and_v(v:after
  ↳ (1893456000),and_v(v:sha256(a84dce40975727c398023c fbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↳ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsYEAaQfKjKwBuU179LAWG5GhJMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLVdPkEC/<0;1>/*,[b8688df1/84'/0'/0']
  ↳ xpub6DNfJehqF1LU9kwaqDu12Ajpz9psYVtbGhTykQo1CYdkkqV2vAyR4D1wXSTTDujWhZvY1AtV6ENGKwGwBLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↳ xpub6DNfJehqF1LUsoh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvKqFDLWuyCivJ4eMDxgFtiTxnazgp8rYvZQSKR3L8EGRS9asBarqpe/<0;1>/*)),or_i(and_v(v:older
  ↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↳ xpub6DNfJehqF1LUwZ1DrFCQ365CDMPLESsrrqpSwaz4eRrtYz9qWdsVH9JqshPQ6yNgdG4VXGKbLgCkKkXtWtB5B4iCuHcmjomqR8z6NNXnpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↳ xpub6DNfJehqF1LUxb8gRSstbR9LcAXwGwD4V678z5FZ7BLftWzWQct3yZb5eW4U8AVjwbKcZ2iegjvKvX2gxpJmBRkk3CE7bz7g4uMV7Qp3TU/<0;1>/*)),and_v(v:older(4255898)
  ↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBbzvudcQg2nS4tH5krm7FGkXl6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpu8p2Xx4wWQK7iGBycAm
  ↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBbzvudcQg2sPDRWpqfEn5VzPiwrd1zNes4aHmnnUog9Jmc2fc2J5fM2E39Yz2iakmpWRpa3rXhZGnm4dGKiJZEXmTCftQpWBGd9ihVks
  ↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBbzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABF4j48TRxUnRd5eGgEXfgP63xcuTV9uny7pQBFbu3KKcN9rNxcNGRADt9BD1gkBGZ
  ↳ /<0;1>/*)))})
```

Validate and canonicalise it. The full canonical descriptor, every xpub in full:

```
$ mnemonic export-wallet --descriptor "$(cat taproot.desc)" --format descriptor --network mainnet
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNTnqr2UHCBEsCoL3eEFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{or_i(
  ↳ and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023c fbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↳ xpub6CatWdiZiodmUeTDp8LT5or8nmbKNcuyvz7WYksVfKKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLOp2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↳ xpub6CatWdiZiodmYVtWLTtEQsAg1H9oo5lBmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZS3KA5dh5Lo2PbubMGxCfZtRthV6S2tqquL9w3HSzcUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↳ xpub6CatWdiZiodmNGqCQxxjGN165QxTU4PwNni9WriJYgYf7VxcmuFosRw3focZLgRDbjDjJbqZhPCTfKcaWmL9BuSw98ybKKJtcHgwWeryy6/<0;1>/*)),and_v(v:after
  ↳ (1893456000),and_v(v:sha256(a84dce40975727c398023c fbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↳ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsYEAaQfKjKwBuU179LAWG5GhJMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLVdPkEC/<0;1>/*,[b8688df1/84'/0'/0']
  ↳ xpub6DNfJehqF1LU9kwaqDu12Ajpz9psYVtbGhTykQo1CYdkkqV2vAyR4D1wXSTTDujWhZvY1AtV6ENGKwGwBLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↳ xpub6DNfJehqF1LUsoh1a4XLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvKqFDLWuyCivJ4eMDxgFtiTxnazgp8rYvZQSKR3L8EGRS9asBarqpe/<0;1>/*)),or_i(and_v(v:older
  ↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↳ xpub6DNfJehqF1LUwZ1DrFCQ365CDMPLESsrrqpSwaz4eRrtYz9qWdsVH9JqshPQ6yNgdG4VXGKbLgCkKkXtWtB5B4iCuHcmjomqR8z6NNXnpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↳ xpub6DNfJehqF1LUxb8gRSstbR9LcAXwGwD4V678z5FZ7BLftWzWQct3yZb5eW4U8AVjwbKcZ2iegjvKvX2gxpJmBRkk3CE7bz7g4uMV7Qp3TU/<0;1>/*)),and_v(v:older(4255898)
  ↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBbzvudcQg2nS4tH5krm7FGkXl6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpu8p2Xx4wWQK7iGBycAm
  ↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBbzvudcQg2sPDRWpqfEn5VzPiwrd1zNes4aHmnnUog9Jmc2fc2J5fM2E39Yz2iakmpWRpa3rXhZGnm4dGKiJZEXmTCftQpWBGd9ihVks
  ↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBbzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABF4j48TRxUnRd5eGgEXfgP63xcuTV9uny7pQBFbu3KKcN9rNxcNGRADt9BD1gkBGZ
  ↳ /<0;1>/*)))})
```

```

↪ (65535),multi_a(2,[b8688df1/84'/0'/2']
↪ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqShpQ6yNGdG4VXGKbLgCkKxTwtB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
↪ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzQcT3yZb5eW4U8AVJwbKcZ2iegjvjKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*),and_v(v:older(4255898)
↪ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvdudcQg2nS4tH5Krm7FGkXL6arWxEYoZ1g1GwBaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpU8p2Xx4wMQK7iGBycAm
↪ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvdudcQg2sPDRWpqfEn5VzPiwrldzNes4aHmNUog9Jmc2fc2J5fM2E39Y2yZiakmpWRpa3rXhzGNmd4GKiJEXzmtCftQpwBGd9ihVks
↪ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvdudcQg2uk9BrxsuxCwj5sYsfPYkPahQfmTVABf4j48TRxUnRd5eGgEXfgPJ63xcuTV9uny7pQBfBu3KXcN9rNxcNGRaDT9BD1gkBGZ
↪ /<0;1>/*)))#nerswx7
note: stdout is watch-only – public keys only, cannot spend

```

mint ([73c5da0a/84'/0'/4']) is the key-path; the two `or_i(...)` blocks are the two script leaves; `after(...)` are the absolute (height/time) locks and `older(...)` the relative (blocks/time) locks -- the same four kinds as section 5.

6.3 Engrave + first address

Every key is distinct, so it engraves (watch-only). Take the `md1` chunks and restore to read the first address (this round-trip also proves the **real internal key** at the trunk reconstructs -- the `v0.55.3` non-NUMS feature). The descriptor file (shown again):

```

$ cat taproot.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmXswr13Gd5aNtNqr2UHCBEsCoL3eEFVaM7n8kY5K4daaP83gWQncmzL3Wzt79mEiLiXZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{or_i(
  ↪ and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↪ xpub6CatWdiZiodmUeTdp8LT5or8nmbKNcuYv7WYksVfKKB4RHwCD3YxuvPEbvqAQY3rAPshWcMLOp2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↪ xpub6CatWdiZiodmYVtWLTtEQAsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3quZSJKASdh5Lo2PbubMGxCFZtRthV65T2quL9w3HSzclUn/<0;1>/*,[73c5da0a/84'/0'/2']
  ↪ xpub6CatWdiZiodmBgCQxxjGN165QoTU4PwNNi9WrjYgYf7VxcmuFxsRw3focZLgRbDjDjJbqZhPCTfkcaWmL9BuSw98ybKJktcHgweryy6/<0;1>/*)),and_v(v:after
  ↪ (1893456000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↪ xpub6CatWdiZiodmHurRokjbyCcxrdTDJgTsyEaaQfKjKwbWUi79LAWG5GhJMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLvdPkEC/<0;1>/*,[b8688df1/84'/0'/0']
  ↪ xpub6DNfJehqF1LU59kwaqDu12Ajzp9psYVtbGhTykQoLCYdkkV2yAR4DiwXSTTDjuWhzVy1AtV6ENGKwgbLWqa4wXMR4ZmdpRjQ8G5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↪ xpub6DNfJehqF1LUsohl1aXLG3yuFY1oCRJd6Bkba97xvW6SwvY71Nc6LvqfDLWuyCivJ4eMDxgftiTxnazgp8rYvZSQKR3L8EGrs9asBarqpe/<0;1>/*))),or_i(and_v(v:older
  ↪ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↪ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqShpQ6yNGdG4VXGKbLgCkKxTwtB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↪ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzQcT3yZb5eW4U8AVJwbKcZ2iegjvjKv2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*),and_v(v:older(4255898)
  ↪ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvdudcQg2nS4tH5Krm7FGkXL6arWxEYoZ1g1GwBaNdWRobcJmxH8KQdjuzTzJtwDveibGd83eGzdGCR6NjSqpU8p2Xx4wMQK7iGBycAm
  ↪ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvdudcQg2sPDRWpqfEn5VzPiwrldzNes4aHmNUog9Jmc2fc2J5fM2E39Y2yZiakmpWRpa3rXhzGNmd4GKiJEXzmtCftQpwBGd9ihVks
  ↪ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvdudcQg2uk9BrxsuxCwj5sYsfPYkPahQfmTVABf4j48TRxUnRd5eGgEXfgPJ63xcuTV9uny7pQBfBu3KXcN9rNxcNGRaDT9BD1gkBGZ
  ↪ /<0;1>/*)))

```

```

$ mnemonic bundle --descriptor-file taproot.desc --network mainnet --json | jq -r ".md1[]" > taproot.md1
note: stdout is watch-only – public keys only, cannot spend

```

```

$ cat taproot.md1
md1f40gpqvz6jzwrh4yypm6jzwx75j3m6jzt802ggrh4yyvaafp9rh4yykw75s802gemcn9wrf4m7m5yt9
md1f40gpqvz2jz2sqrqgy2efntvqq7sjqfntk5ymnjqatj0sucqg70h4gdtkemj3rwr4fp6rc6cnxqh4hnx7rv
md1f40gpqvzskmmq5mngy26gz2zgv6mwrda3qzv6a4pxuusyhtunu8xqz8naa2r2akwukvgm4g7w7kdyaf805mf
md1f40gpqvz6gws7xx9k7c9xu6pzkjgqj9vefntsqpplllyqshsnxhqgrcf5gqzn2cvasqu79cwukltfwjcnckn
md1f40gpvpgr9pw0za5z3883w6pgmncdq53eutks2twrg3hckhp5gmutms6ydx9cdxzlry5xgklamn3a93x43x
md1f40gpvpv9sq5x52ryzrt9p9jqpskfufq44hccjz40ltnk8tdswlxxk6gs50wsf6dvu4jev3eqgy8f0nxdna6x
md1f40gpvpv56gl4vs3t2sqwftfh7sept4g4hf5939sedkqhey4nnc0chj6t2x82xgqygmpkv7cuw0m7akf723
md1f40gpvpv622ws2kgdemj2cd8zwj2skzx2wq0q70l4q99vdyh5x08v4yslsppq7nzlkkvymsvlrcugrdkju72
md1f40gpvpz9fkac3cxtqemqkacpcp59u7xtux3d5d847ws4xev6setz7wcgwg0mt0jarageqgfesknkmcceqj
md1f40gpvpz0ptr0vx3c9895ghka0gk58rcw6qzw4cdxzd3en9mnhxqdpws2ec6jxd33axqmwg8tzlv563mv
md1f40gpvznmjszkn4t8pc6l88xna4nqjuhuhc32y6rxwttx8w3uxf33474jxxpl30507ttsmq84l460gwc0g4w
md1f40gpvzcx7rcsyrzx48lzn7wute526tvfexawdfdy4vy7y7s9c84gwagyhhf8ugmwft46ekqd54ka5c8c48mx
md1f40gpvr0a7t0c6gn7uss8mxrkma6e6mkqp24zvzxnc1w0xxzg0sgstqtmmy87px3u9mlm3vgd8s0y6hrrvud
md1f40gpvrvmnj3aafe7gyny7hjyqsfhjzfaf5hs5jwzh3mhpgjm2ctzuevmwdfap2f5z19lsqqkg20r8nj9q
md1f40gpvrkdtfh37zrqnvhxzcK5fhfadj2Lxnax2ywfue5grx4rgemgh0nk5eg0ze8tj0qfnr3syntujt2jgxs5pl
md1f40gpvr6zg55fzuu4z339d06w5s99f4v94tc38peqc5jysp6pp3g08exa4srx0889h3ddqqnqhd12vm2eucf
md1f40gpvpv6ttmcl1y57acv6els0u3jj86dd5pe7m355rtwt8r3a2s45tszcmrremst4u6j4atqccq4mtqrtm0ny
md1f40gpvyg6uhd0m3m5l9lvu9cde6pcswdrn73ejs4lts9qcszq6fnt9u8jvs93uep30g6dq0k3hj4xq9v7xn
md1f40gpvynng0aj0c55p20eykxhftlk456pzlvcqh3za8msktlmcqvgkmuJ2dwpkhhlhm290qceu48zwu553kp
md1f40gpvyckdgr4nrpxszs4jyqyut3ee64sfnz6ryr348le5ljx4sv0e02cjfm14zmy9g36c6spmukzfw55f
md1f40gpv9rfpgzjjzpmz7nhnk6yh5veeegnzkfxcgdg3sypfknoRu6gc370szpx6rudkq2e7xecrr560mc7uy987
md1f40gpv9tzzetsg6tpqua7cty0t4c4w6rtijnmht8891lvhne3ktx5etuhL2rmxpaqj6ey0lg6d4mp519hjrne
md1f40gpv95xj46th52vv3c3kju0767f9u9j0ynrejo0uqenhu2pdakxrrtfe5qck40z57z8ylqssej5h2v929gq
md1f40gpv9eyh2tqvnhxzcK5fhfadj2Lxnax2ywfue5grx4rgemgh0nk5eg0ze8tj0qfnr3syntujt2jgxs5pl
md1f40gpvx92k0c250h62ypscarml7tgutcg8uf4m7gfc8l1fv33knpwrthl06am5gdw9a9nqfkhq0hnyzweea

```

```

$ mnemonic restore --network mainnet $(sed "s/~/-md1 /" taproot.md1)
miniscript policy restore (12 cosigners)
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: tr([73c5da0a/84'/0'/4']xpub661MyMwAQrbCYUkSqsBgaz1Lob8pCa1rM1SJBCEGcYcYP9LisxZ2m1GoqDj137XvHdY2NkctqiElixaAFqYHf91CFpFpKicVb7TzvrGSe
  ↪ /<0;1>/*,{or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a
  ↪ /84'/0'/0']xpub661MyMwAQrbCFHwVpCiBTxd2Caj7vZnHFHJ5G59Aue2YkXsRz5q9Ga0rRjJvHHTZtCiHWSZgm55beaaTHwMhpGc7SMdqMHRXZi8as/<0;1>/*,[73c5da0a
  ↪ /84'/0'/1']xpub661MyMwAQrbC6axoYcLaxHHXZgEgSRQmN2P5ung8M3MNE535mLuq7zjnrmKYA5eX6ehicVbU1FFPU39LGxbY8PmPLQxVRQMPFa3Q7spa/<0;1>/*,[73c5da0a
  ↪ /84'/0'/2']xpub661MyMwAQrbC6uXAHBK3oquZS1HJiz2fVZ2idNC4G6LGTJyGZgPKP7fv1N6euv5GzY18JD3WBG3SoLat23TLAVjhQMxdVMAqymNHg3RFT8/<0;1>/*))),and_v(v:
  ↪ after(1893456000),and_v(v:sha256(a84dce40975727c398023cfd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↪ xpub661MyMwAQrbCGLCZcljg25oG8w5yqS5EXNM9uMks6vrrh4pRDC8UAmaynovThuKraidMeEKJ2fCqbw1eF76aeu1vrGtLXUjIXr4r9N1TQZ/<0;1>/*,[b8688df1/84'/0'/0']
  ↪ xpub661MyMwAQrbC6owNgeNcL58Cgl2vnZybpJqtKbCmSQMdqZqcDwqq3CXGx75BqvcNC5ANuW6n1soN7JFK2j3HfxV57Nm2RLKo2UzHgbs6/<0;1>/*,[b8688df1/84'/0'/1']

```

```

    ↪ xpub661MyMwAQbRcEv3U8uuxavsQa8LNNYwcNge8rT7SaMS558KiEwXoP72TQ8ARYjczPTtVQz6CxcabTEE3XchmYvSiHcVbC9h17CmyfG7sVq9/<0;1>/*))) ,or_i(and_v(v:older
    ↪ (65535),multi_a(2,[b8688df1/84'/0'/2'])
    ↪ xpub661MyMwAQbRcG7Xht9EwgNucA47Rmgg8Bn5bNmFdJmKotHQDXirpogQHkVNRcAy6KwGnUYUMNBFcNaRq4WnsqWw2VNuddW6ymHXfVpk4c3/<0;1>/*,[b8688df1/84'/0'/3'])
    ↪ xpub661MyMwAQbRcEbqBvNKLdUtudGA2PHAbtWuUHKe3CKjZCaLxLjGS68SjpwBCnXsj8xPGXaV9ZWL3j9ktbed8y1aeNVK95HrkGfHfGBXMS5EH/<0;1>/*))) ,and_v(v:older(4255898)
    ↪ ,multi_a(1,[28645006/84'/0'/0']xpub661MyMwAQbRcEdBoFbaBgbnse74WruYebXRsSmzq8jztzhtDnXTV2yNQPzgs3ubwuNp7yrSHnECa5xHgnEDH4HSGWqLTYdi6nWZCfXPk
    ↪ /<0;1>/*,[28645006/84'/0'/1']xpub661MyMwAQbRcH2WNmbt34pZ8WdtpxdYoe6E4r5o8rXedve17srmalLCEjM8WcpV6k7xsc36KpBntYUdqo5dpcFMzRfzS2Sa4C5DRty4eDNF
    ↪ /<0;1>/*,[28645006/84'/0'/2']xpub661MyMwAQbRcGqcAAnB9mhvQsdUx2fKasUoXT2fMpt2tF294wRfAkhuLhZUJkjQ5pgnd9Ny9EwrgcHbAASVnQShCbfnhGskAk2k6yGowXAv
    ↪ /<0;1>/*)))#7cy3x3q9
first recv: bc1p9stcwz5597fmkxae9343k8edzkcvdcz9qp65r6p447pg0et82yqst3d2c
cosigner @0: 73c5da0a [84'/0'/4'] from md1 (not independently verified)
cosigner @1: 73c5da0a [84'/0'/0'] from md1 (not independently verified)
cosigner @2: 73c5da0a [84'/0'/1'] from md1 (not independently verified)
cosigner @3: 73c5da0a [84'/0'/2'] from md1 (not independently verified)
cosigner @4: 73c5da0a [84'/0'/3'] from md1 (not independently verified)
cosigner @5: b8688df1 [84'/0'/0'] from md1 (not independently verified)
cosigner @6: b8688df1 [84'/0'/1'] from md1 (not independently verified)
cosigner @7: b8688df1 [84'/0'/2'] from md1 (not independently verified)
cosigner @8: b8688df1 [84'/0'/3'] from md1 (not independently verified)
cosigner @9: 28645006 [84'/0'/0'] from md1 (not independently verified)
cosigner @10: 28645006 [84'/0'/1'] from md1 (not independently verified)
cosigner @11: 28645006 [84'/0'/2'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only – public keys only, cannot spend

```

(Restore re-serialises each key as a depth-0 xpub661My... -- a different descriptor string, identical addresses; that is how the md1 card stores keys.)

6.4 Export for wallets (Nunchuk / Core / Sparrow)

descriptor and bitcoin-core both work for taproot. The descriptor file (shown again):

```

$ cat taproot.desc
tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNtNqr2UHCBEsCoL3eEFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,[or_i(
    ↪ and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0'])
    ↪ xpub6CatWdiZiodmUeTdp8LT5or8nmbKNcuyvz7WYksVFkKB4RHwCD3XyuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1'])
    ↪ xpub6CatWdiZiodmYVtWltEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEv3quZSJKASdh5Lo2PbubMGxCfZtRthV6ST2qquL9w3H5zcUn/<0;1>/*,[73c5da0a/84'/0'/2'])
    ↪ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNNi9WrijYgYf7VxcmuFxoSrw3focZLgRdbjDjJbqZhPCTfkcaWmL9BuSw98ybKKJtcHgwWeryy6/<0;1>/*))) ,and_v(v:after
    ↪ (1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3'])
    ↪ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsYEAaQfKjKwBwUi79LAWG5gHjMCQB7BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLVPkEC/<0;1>/*,[b8688df1/84'/0'/0'])
    ↪ xpub6DNfJehqF1LU59kwaQDu12AjPz9psYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDjWhZVy1AtV6ENGKwgbLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1'])
    ↪ xpub6DNfJehqF1LUsoh1a4XLG3yuF1oCRJd6Bkba97xvW6SwvY71Nc6LvKqFDLWuyCivJ4eMDxgFtiTxnazgp8rYvZQSKR3L8EGRs9asBarqpe/<0;1>/*))) ,or_i(and_v(v:older
    ↪ (65535),multi_a(2,[b8688df1/84'/0'/2'])
    ↪ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqShpQ6yNgdG4VXGKbLgxcKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3'])
    ↪ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzWQct3yZb5eW4U8AVJwbKcZ2iegjvkvX2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*))) ,and_v(v:older(4255898)
    ↪ ,multi_a(1,[28645006/84'/0'/0']xpub6DBbzvudcQg2nS4tH5Krm7FGkXL6arWxEyoZ1g1GwbaNdWRobcJmxH8KQdjuzTJ2twDveibGd83eGzdGCR6NjSqpU8p2Xx4wWQK7iGBycAm
    ↪ /<0;1>/*,[28645006/84'/0'/1']xpub6DBbzvudcQg2sPDRWpqfEn5VzPiwrldzNes4aHmNUog9Jmc2fc2J5fM2E39Yz2iakmpWRpa3rXhZGnm4GK1JEZxmTctfQpwBgD9ihVks
    ↪ /<0;1>/*,[28645006/84'/0'/2']xpub6DBbzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABFJ4j8TRXUnRd5eGgEXfgPJ63xcuTVun9y7pQBfBu3XKCN9rNxcNGRaDT9BD1gkBGZ
    ↪ /<0;1>/*)))#7cy3x3q9

```

Bitcoin Core `importdescriptors` payload:

```

$ mnemonic export-wallet --descriptor "$(cat taproot.desc)" --format bitcoin-core --network mainnet
[
  {
    "active": true,
    "desc": "tr([73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNtNqr2UHCBEsCoL3eEFvAm7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{
      or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0'])
      ↪ xpub6CatWdiZiodmUeTdp8LT5or8nmbKNcuyvz7WYksVFkKB4RHwCD3XyuvPEbvqAQY3rAPshwCMLoP2fMFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1'])
      ↪ xpub6CatWdiZiodmYVtWltEQsAg1H9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEv3quZSJKASdh5Lo2PbubMGxCfZtRthV6ST2qquL9w3H5zcUn/<0;1>/*,[73c5da0a/84'/0'/2'])
      ↪ xpub6CatWdiZiodmNGqcQxxjGN165QxTU4PwNNi9WrijYgYf7VxcmuFxoSrw3focZLgRdbjDjJbqZhPCTfkcaWmL9BuSw98ybKKJtcHgwWeryy6/<0;1>/*))) ,and_v(v:after(1893456000)
      ↪ ,and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3'])
      ↪ xpub6CatWdiZiodmHurRokjbycCrxddTDJgTsYEAaQfKjKwBwUi79LAWG5gHjMCQB7BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLVPkEC/<0;1>/*,[b8688df1/84'/0'/0'])
      ↪ xpub6DNfJehqF1LU59kwaQDu12AjPz9psYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDjWhZVy1AtV6ENGKwgbLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1'])
      ↪ xpub6DNfJehqF1LUsoh1a4XLG3yuF1oCRJd6Bkba97xvW6SwvY71Nc6LvKqFDLWuyCivJ4eMDxgFtiTxnazgp8rYvZQSKR3L8EGRs9asBarqpe/<0;1>/*))) ,or_i(and_v(v:older
      ↪ (65535),multi_a(2,[b8688df1/84'/0'/2'])
      ↪ xpub6DNfJehqF1LUwZ1DrFcQN365CDMPLSsrrqpSwaz4eRrtYz9qWdsVH9JqShpQ6yNgdG4VXGKbLgxcKkXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3'])
      ↪ xpub6DNfJehqF1LUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzWQct3yZb5eW4U8AVJwbKcZ2iegjvkvX2xggJpMBRk3CE7bz7g4uMV7Qp3TU/<0;1>/*))) ,and_v(v:older(4255898) ,
      ↪ multi_a(1,[28645006/84'/0'/0']xpub6DBbzvudcQg2nS4tH5Krm7FGkXL6arWxEyoZ1g1GwbaNdWRobcJmxH8KQdjuzTJ2twDveibGd83eGzdGCR6NjSqpU8p2Xx4wWQK7iGBycAm
      ↪ /<0;1>/*,[28645006/84'/0'/1']xpub6DBbzvudcQg2sPDRWpqfEn5VzPiwrldzNes4aHmNUog9Jmc2fc2J5fM2E39Yz2iakmpWRpa3rXhZGnm4GK1JEZxmTctfQpwBgD9ihVks
      ↪ /<0;1>/*,[28645006/84'/0'/2']xpub6DBbzvudcQg2uk9BrxsuxCwjsYsbfPYkPahQfmTVABFJ4j8TRXUnRd5eGgEXfgPJ63xcuTVun9y7pQBfBu3XKCN9rNxcNGRaDT9BD1gkBGZ/<0;1>
      ↪ ))#q3s6u2uk",
    "internal": false,
    "range": [
      0,
      999
    ],
    "timestamp": 0
  },
  {
    "active": true,

```



```
$ mnemonic compare-cost --miniscript "and_v(v:pk(F),ripemd160(06d05e2f02fb90ddf98d8cd95d806ba12b27aff4))"
Input: and_v(v:pk(F),ripemd160(06d05e2f02fb90ddf98d8cd95d806ba12b27aff4))
Wrapper comparison: wsh(M) vs tr(NUMS, {M})
Feerate: 1.0 sat/vB
```

Condition	wsh vB	tr vB	Δ vB	wsh sats	tr sats	Δ sats
F + preimage(h0)	68	90	+22	68	90	+22

note: per-condition vbytes are rounded individually; absolute numbers may differ by ±1 from real-tx accounting, Δ values are correct
note: input contains hash-preimage fragments; preimage-known rows are enumerated assuming the user can supply each preimage (cost only – no preimage
↳ knowledge is implied)

compare-cost models each input as a single leaf at the tree root (depth-0, 33-byte control block); add the real Merkle depth (+8 vB per level):

Placement	revealed script	tr vB (depth-0)	real depth	+ control block	real tr
Folded into	whole	139	1 (still 2	+8	~147 vB
Leaf B	or_i(t3,t4,t5)		leaves)		
Dedicated leaf	just t5	90	2 (now 3	+16	~106 vB
			leaves)		

A dedicated leaf is ~41 vB (~28%) cheaper to spend the new condition: not revealing tiers 3+4 (~49 vB) beats the +8 vB for the extra depth level. It is better still in practice because folding also makes *t3* and *t4* spends reveal *t5*'s script, and a 3-leaf tree can place the **most-used** condition in the shallow slot and bury cold paths deeper.

Rule of thumb: in taproot, one-condition-per-leaf almost always wins -- each sibling script you avoid revealing is worth far more than the 8 vB/level of depth -- and you order leaves hot-shallow / cold-deep.

The catch: cost says depth-2, but master cannot build depth->=2 yet (the PR-#953 taptree-Display bug, 6.1), so today the only buildable option is the folded depth-1 leaf at the ~28% premium. That premium is the concrete, quantified motive for landing the upstream fix.

6.6 Taproot multisig (NUMS), cross-checked against Bitcoin Core

The taproot wallets above are deliberately complex. The *simplest* taproot **multisig** is a single `sortedmulti_a` 2-of-3 leaf under the BIP-341 **NUMS** ("nothing-up-my-sleeve") internal key -- an **unspendable** key-path, so the only way to spend is the sorted multisig script. It round-trips on the **shipped** binary, and Bitcoin Core derives the **identical** first address. (The toolkit's own `tests/bitcoind_differential.rs` gate proves exactly this -- `bundle -> restore -> first_addresses` VS `Core deriveaddresses` -- for the taproot-multisig corpus against a pinned Bitcoin Core v27.0.) Reuse the three section-3 cosigners:

```
$ cat taproot-multi.desc
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,sortedmulti_a(2,[73c5da0a/87'/0'/0']
  ↳ xpub60BjiYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub6CbhrPzY2z7NcCGGjLAJLq8iRyjUfwmdXQs66MxTVURekQb9DpLnVJ5D1qpatZjUuPGTyxf5TYUlvA34YFE9FHB4TvfYmokYLVsyEFZFt9/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub6DB7HNqw6CZojxN85NuFTPWZhi2FagSnextPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hJhN2uazz4z/<0;1>/*))
```

Validate and canonicalise it (the NUMS hex is the BIP-341 unspendable H-point):

```
$ mnemonic export-wallet --descriptor "$(cat taproot-multi.desc)" --format descriptor --network mainnet
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,sortedmulti_a(2,[73c5da0a/87'/0'/0']
  ↳ xpub60BjiYnc4ewKti13Q1L35bqdodw5z3VGJnf516B3icHrEGEUcCuG5GVQDZtH8Xmsyt3Fs9YDNwLaqjUbbRidwXZ6sxufZcr4VqqzrXvicM/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub6CbhrPzY2z7NcCGGjLAJLq8iRyjUfwmdXQs66MxTVURekQb9DpLnVJ5D1qpatZjUuPGTyxf5TYUlvA34YFE9FHB4TvfYmokYLVsyEFZFt9/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub6DB7HNqw6CZojxN85NuFTPWZhi2FagSnextPS1rv3nYQhngkmdHgb7iebYvTFmFKKDA3ozf5yezDsCH6cXAw3WZijviSZtZC2hJhN2uazz4z/<0;1>/*))#8nz0lwja
note: stdout is watch-only – public keys only, cannot spend
```

Engrave the watch-only card set and read the first address back from the md1 chunks alone:

```
$ mnemonic bundle --descriptor-file taproot-multi.desc --network mainnet --json | jq -r ".md1[]" > taproot-multi.md1
note: stdout is watch-only – public keys only, cannot spend
```

```
$ mnemonic restore --network mainnet $(sed "s/^/--md1 /" taproot-multi.md1)
2-of-3 multisig restore
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,sortedmulti_a(2,[73c5da0a/87'/0'/0']
  ↳ xpub661MyMwAQrbCFrooZ2966EcDmVX5MoFXZhuJqXTudvJzwbTBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPwDUSNjsY5JsgVBQoD6KrbdCLL/<0;1>/*,[b8688df1/87'/0'/0']
  ↳ xpub661MyMwAQrbCEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXYukNGMK2XFRbAaCLYEJDUufNoBxTNa68i5MYhqmrEkfhjzgHCUEcvJBhXS5bk4RW/<0;1>/*,[28645006/87'/0'/0']
  ↳ xpub661MyMwAQrbCEdy4jr5EtEHQbctfscE6a99DGLr2cW4HnnmBsXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/<0;1>/*))#k0lsap8u
first recv: bc1p550zvnachy40z6hh8llka93mkm0c3635smp264ck6rfd0dc8s00n8c8
cosigner @0: 73c5da0a [87'/0'/0'] from md1 (not independently verified)
cosigner @1: b8688df1 [87'/0'/0'] from md1 (not independently verified)
cosigner @2: 28645006 [87'/0'/0'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only – public keys only, cannot spend
```

`restore` reports a `bc1p...` Taproot address. Confirm it against Bitcoin Core's **independent C++** derivation: `deriveaddresses` on the receive (`.../0/*`) descriptor (split from the `<0;1>` multipath, which Core rejects):

```
$ bitcoin-cli -chain=main deriveaddresses "tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,sortedmulti_a(2,[73c5da0a/87'/0'/0']
  ↳ xpub661MyMwAQrbCFrooZ2966EcDmVX5MoFXZhuJqXTudvJzwbTBfPQSc5JzX52fvS18oqSdEJXJ4kTGRJ76wPwDUSNjsY5JsgVBQoD6KrbdCLL/0/*,[b8688df1/87'/0'/0']
  ↳ xpub661MyMwAQrbCEnFgxHRLx7i1fnjcBPgc71qy8mVkbGXYukNGMK2XFRbAaCLYEJDUufNoBxTNa68i5MYhqmrEkfhjzgHCUEcvJBhXS5bk4RW/0/*,[28645006/87'/0'/0']
  ↳ xpub661MyMwAQrbCEdy4jr5EtEHQbctfscE6a99DGLr2cW4HnnmBsXDoe3odGzRiw3hcRM5wfKcQmb7s5FjdGrR6SrExXmeopaoY9Lk7tQusDjN/0/*))#mk8v dqmt" "[0,0]"
["bc1p550zvnachy40z6hh8llka93mkm0c3635smp264ck6rfd0dc8s00n8c8"]
```

Byte-for-byte the same `bc1p...` that `restore` reported -- the toolkit's own derivation (which v0.49.1 routes *around* the codec for taproot) agrees with Bitcoin Core. `tr(NUMS,sortedmulti_a)` is a shape **only** the toolkit can render -- its pinned rust-miniscript fork has `sortedmulti_a`, the codec's crates.io build does not -- so this end-to-end check is the only place an external oracle sees it.

Appendix A -- the public test seeds used

Slot	BIP-39 test phrase	Fingerprint (BIP-87)
@0	abandon abandon abandon abandon abandon abandon ↳ abandon abandon abandon abandon abandon about	73c5da0a
@1	legal winner thank year wave sausage worth ↳ useful legal winner thank yellow	b8688df1
@2	letter advice cage absurd amount doctor acoustic ↳ avoid letter advice cage above	28645006

These are world-known BIP-39 vectors with no funds. Sections 5-6 derive their keys from these same three seeds at distinct `m/84'/0'/N'` accounts. Generated with `mnemonic v0.55.3` on Linux. See the in-repo manual (`docs/manual/`) for the authoritative per-flag reference.

Appendix B -- EXPERIMENTAL: depth->=2 taproot reconstruction (NOT FOR REAL FUNDS)

EXPERIMENTAL -- DO NOT USE FOR REAL FUNDS. Everything in this appendix uses mnemonic `depth2`, a proof-of-concept binary built from the never-merged branch `experimental/taproot-depth-ge2`. It pins an **unreleased** rust-miniscript commit (the PR-#953 merge, in no crates.io release) to lift the `depth->=2` taptree cap. It is **not** part of any shipped release, is not on the install path, and must never be used to secure real funds. The shipped mnemonic v0.55.3 documented everywhere else in this guide deliberately **refuses** `depth->=2` (sections 6.1 and below); that refusal is the supported behavior until a rust-miniscript release > 13.1.0 ships #953.

Master refuses to *restore* a `depth->=2` taptree because the shipped miniscript pin mis-formats nested taptrees. The POC bumps that pin and lifts the cap. Build it under a **distinct name** so it never overwrites your real mnemonic (both report the same version string 0.55.3, so the command name is the only thing that tells them apart):

```
$ git clone https://github.com/bg002h/mnemonic-toolkit && cd mnemonic-toolkit
$ git checkout experimental/taproot-depth-ge2
$ cargo build --release --manifest-path crates/mnemonic-toolkit/Cargo.toml
$ cp target/release/mnemonic ~/.cargo/bin/mnemonic-depth2 # DISTINCT name
```

Same version string, different binaries -- tell them apart by command name:

```
$ mnemonic --version
mnemonic 0.55.3
```

```
$ mnemonic-depth2 --version
mnemonic 0.55.3
```

Recall the `depth-2` four-leaf descriptor from section 6.1 (one tier per leaf):

```
$ cat taproot-4leaf.desc
tr{[73c5da0a/84'/0'/4']xpub6CatWdiZiodmeXswr13Gd5aNTNqr2UHCBCesCol3eEFvA7n8kY5kS4daaP83gWQncmzL3Wzt79mEiLiX6XZs6XQmGcQNeQ4HcjfVTn9TuXE/<0;1>/*,{and_v(v:
  ↳ after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a/84'/0'/0']
  ↳ xpub6CatWdiZiodmUeTDp8LT5or8nmbKNcuyvz7WYksVFkKB4RHwCD3XyuvPEbvqAQY3rAPshWcMLOP2fvmFMKHPJ4ZeZXYVUHLv1VMrjPC7PW6V/<0;1>/*,[73c5da0a/84'/0'/1']
  ↳ xpub6CatWdiZiodmYVtWLTQsAgH9ooS1bmsJUBwQ83FE1Fyk386FWcyicJgEzV3qz5JKA5dh5Lo2PbubMGxCfZTrthV6ST2qquL9w3H5zscU/<0;1>/*,[73c5da0a/84'/0'/2']
  ↳ xpub6CatWdiZiodmbNGcqQxxjGN165QxTU4PwNNi9WrijYgYf7VxcmuFxsRw3focZLgRDbjDjJbqZhPCTfkcaWmL9BuSw98ybKKJtcHgWeryy6/<0;1>/*)),and_v(v:after
  ↳ (1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
  ↳ xpub6CatWdiZiodmHurrRokjbyccrXddTDJgTsyEaaQfKjkwBwUi79LAWG5gHjMCQ87BeJc47MkubXuZdf45JZHk1qcr1GZ5EwREUDVDLVdpKEC/<0;1>/*,[b8688df1/84'/0'/0']
  ↳ xpub6DNfJehqFILLUs9kwaqDu12Ajzp9psYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDujWHZVy1AtV6ENGKwGwBLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*,[b8688df1/84'/0'/1']
  ↳ xpub6DNfJehqFILLUs9kwaqDu12Ajzp9psYVtbGhTykQo1CYdkkqV2vAyR4DiWXSTTDujWHZVy1AtV6ENGKwGwBLWqa4wXMR4ZmdpRjQBG5EgTV/<0;1>/*)),{and_v(v:older
  ↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
  ↳ xpub6DNfJehqFILLUwZ1DrFCQn365CDMPLSsrrqpSwaz4eRtYz9qWdsVH9JqshPQ6yNgdG4VXGKbLgCkKXtwTB5B4iCuHcmjomqR8z6NNxNpU51/<0;1>/*,[b8688df1/84'/0'/3']
  ↳ xpub6DNfJehqFILLUxb8gRSstbR9LcAXWgWd4V678z5FZ7BLftWzWcQt3yZb5eW4U8AVJwbKcZ2ieggjvKv2xggJpMBRkK3CE7bz7g4uMV7Qp3TU/<0;1>/*)),and_v(v:older(4255898)
  ↳ ,multi_a(1,[28645006/84'/0'/0']xpub6DBBzvudcQg2nS4tH5Kmr7F6KXL6arWxEYoZ1g1GwbaNdWRobcJmxH8KQdjuZTJ3twDveibGd83eZdGCR6NjSqpu8p2Xx4wWQK7iGBycAm
  ↳ /<0;1>/*,[28645006/84'/0'/1']xpub6DBBzvudcQg2sPDRWpqqfEn5VzPwrd1zNes4aHmMNUog9Jmc2fC2J5FM2E39Yz2iaKmpWRpa3rXhzGNmd4GKiJEZxmTCftQpWBGd9ihVks
  ↳ /<0;1>/*,[28645006/84'/0'/2']xpub6DBBzvudcQg2uk9BxrsuxCwjsYsbfPYkPahQfmTVABfJ4j8TRXUnR5eGgEXfgP363xcuTVUn9y7pQBFb3XKCNrNxcNGRaDT9BD1gkBGZ
  ↳ /<0;1>/*))}}
```

The engraved card is a faithful backup even on master -- `bundle` accepts it; only *reading it back* is gated. Make the card and pull its md1 chunks with the shipped binary:

```
$ mnemonic bundle --descriptor-file taproot-4leaf.desc --network mainnet --json | jq -r ".md1[]" > depth2.md1
note: stdout is watch-only - public keys only, cannot spend
```

```
$ cat depth2.md1
md1f8eucvq9zem6jzwrh4yypm6jzxw75sj3m6jzt802ggrh4yyvaafp9rh4yykw75s802ggemcsk65ef96dmuhe
md1f8eucvq2jz2sqrqgy29fntvq75jgfnktSymnjqjatj0sucqg70h4gdtkemje3rw4fp6rc6cqzhan8ac8kjiyw
md1f8eucvqsckmmq5mngy26gzzg6v6mwrda3qzv6a4pxuusydh2unu8xqz8naa2r2akwukvgm4gaa2pacrd4qv84
md1f8eucvq6gws7xx9k7c9xu6pzkjqqj9v9fntsqpllllyqshsnxhqggrcf5gqzn2cvasqu79capl4rdda7hr0z
md1f8eucvprg9pw0za5z3883w6pgmncdq53euts2twrg3hckhp5gmuts6ydx79cdzxly95xg4vfvdrdp8fdd6
md1f8eucvpg5qx52ry2qrt9pj9qpskufqq44hccjz40ltnk8tdswlxxk6gs50wsf6dvu4jev3egrhnl7l0chmf9x6
md1f8eucvp56gl4v53t2qswfth67hsept4g4hf5939sedkqhey4nncochj6t2x82xgqygmppkgatgel8geqxxkd
md1f8eucvp622ws2kgdemj2cd8zuj2skzx2wq0q70l4q99vdyh5x0z8v4yslsp8q7nzlkkvms0vh0v5afqayzk
md1f8eucvz9fka3cxteqm6kacqcp59u7xtux3d5d847wsf4xev6setz7wccgwmmt0jaragegr6d8x0glwkquw
md1f8eucvz9p0tr0vxf3c9895ghkha0gk58rcw6qzw4cdxzd3en9mnxhdpwpx2ec6jxdf33axqcausm7pgz4f8s
md1f8eucvznmj3zkn4t8pc6l88xna4nxjquuhc32y6rwxxttx8w3xuf33474jxxpl30507ttsmqxytzn2nkucsfq
md1f8eucvzc7crsyrzx48lzn7wute526tvfexawdfdy4vy7y7s9c84gwagyhfh8ugmfft46ekq8ppdgxr6l86
md1f8eucvrr0a7t0c6gn7uss8mxrkma6emkqp24vzxncw0xxzg0gsgtmmwy87px3u9mlm3vgw5yc5xf4g4v5q3
md1f8eucvrvnm3jaafe7gyny7hjgysfhfzafaf5hs5jwzh3mhpjgm2ct2uevmwdfafap2f5zL9lsrnzln6ax3u2eu
md1f8eucvrkdtfh372tg9k3d35qm006q30450k4vmqvakpyw77pkdanhr5g2a605szqzpggggn3cjkltnejgh6
md1f8eucvr6zg55fzuu4z339d06w5599f4v94t38peqc5jysp6pp3g08exa4srx0889h3ddqqsnnr60kluky44
md1f8eucvpy6tmcly57ac6vels0u3jj86dd5pe7m355rttw8r3a2s45tsczmmremst4u6j4atqmt5zth78a5h0c
```

```
md1f8eucvgy6uhd0m3m5l9lvu9cde6pcswdr73e5s4lts9qcszq6fnt9u8jvs93uep30g6dqqv99qzfcynrx60
md1f8eucvynng0aj0c55p20eykxhftlk456pzlvcqh3za8mskltmcqvgkmuj2dwpkhhlhm290qc60psjjscz7v2a
md1f8eucvyckgdgr4nrpzxs24jyqytut3ee64sfz6ryr348le5ljx4sv0e02cjfml4zmy9g36cer4vv2u2lmgk4
md1f8eucv9rfpgzjjzpmz7nhnk6yh5veegnzkfxcgdg3sypfkn0ru6gc370szpx6rudkq27excsqdl8x62tamz
md1f8eucv9tzzetsgt6pqua7cty0t4c4w6rtjnmht889ltvhne3ktx5etuhl2rmxpaqj6ey0lge7pv3gpppm09
md1f8eucv95xj46th52vv3c3kj0u767gfu9j0ynrej00uqenhu2pdakxrqtfe5qck40z57z8ylqnr9yt5gn9a5u
md1f8eucv9eyh2tqvnhxzckz5kf hacj7d2lxxax2ywfue5grx4rgemhg0nk5eg0ze8tj0qfnr3s8qltzh5k7fvar
md1f8eucvx92k0c250h62ypsarcmrl7tgutcg8uf4m7gfc8lfv33knprwthl06am5gdw9aqnfqp9rhlf0fq5pp9p
```

The shipped `mnemonic` refuses to restore it (the supported behavior -- the card stays a faithful backup):

```
$ mnemonic restore --network mainnet $(sed "s/^-md1 /" depth2.md1)
error: taproot tree depth ≥ 2 (3 leaves) is not yet restorable – the pinned miniscript mis-prints nested taptrees (FOLLOWUP upstream-miniscript-taptree-
↳ depth2-display-asymmetry); the engraved card remains a faithful backup
```

The experimental `mnemonic-depth2` reconstructs the very same card -- depth-2 taptree and all -- and prints a loud EXPERIMENTAL advisory on stderr before the reconstructed descriptor and first address:

```
$ mnemonic-depth2 restore --network mainnet $(sed "s/^-md1 /" depth2.md1)
EXPERIMENTAL: depth≥2 taproot reconstruction relies on an UNRELEASED rust-miniscript commit (#953, in no crates.io release) – proof-of-concept only; do
↳ NOT use for real funds and do NOT merge. Rebuild when miniscript > 13.1.0 ships.
miniscript policy restore (12 cosigners)
CONFIRM: verify each cosigner fingerprint against your records before importing.
descriptor: tr([73c5da0a/84'/0'/4']xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ /<0;/>1/*,{and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(3,[73c5da0a
↳ /84'/0'/0']xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ /84'/0'/1']xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ /84'/0'/2']xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi_a(2,[73c5da0a/84'/0'/3']
↳ xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ (65535),multi_a(2,[b8688df1/84'/0'/2']
↳ xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ ,multi_a(1,[28645006/84'/0'/0']xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ /<0;/>1/*,[28645006/84'/0'/1']xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ /<0;/>1/*,[28645006/84'/0'/2']xpub661MyMwAQrbCEyUKSqsBgaz1Lob8pCa1rM1Sj8CEZGcYyP9LisxZ2m1goDqj137XvHdY2nNkctqiElixaFqYHf91CFpFpKicVb7TzvrGsE
↳ /<0;/>1/*)}#5trrrgdg0
first recv: bc1p6yc7kzttsafpr6hrrhsaefuyqxvee4j48zdrqt4kl9ers68mhestwn66
cosigner @0: 73c5da0a [84'/0'/4'] from md1 (not independently verified)
cosigner @1: 73c5da0a [84'/0'/0'] from md1 (not independently verified)
cosigner @2: 73c5da0a [84'/0'/1'] from md1 (not independently verified)
cosigner @3: 73c5da0a [84'/0'/2'] from md1 (not independently verified)
cosigner @4: 73c5da0a [84'/0'/3'] from md1 (not independently verified)
cosigner @5: b8688df1 [84'/0'/0'] from md1 (not independently verified)
cosigner @6: b8688df1 [84'/0'/1'] from md1 (not independently verified)
cosigner @7: b8688df1 [84'/0'/2'] from md1 (not independently verified)
cosigner @8: b8688df1 [84'/0'/3'] from md1 (not independently verified)
cosigner @9: 28645006 [84'/0'/0'] from md1 (not independently verified)
cosigner @10: 28645006 [84'/0'/1'] from md1 (not independently verified)
cosigner @11: 28645006 [84'/0'/2'] from md1 (not independently verified)
UNVERIFIED: no --from/--cosigner cross-check supplied; verify each cosigner fingerprint above against your records before importing
note: stdout is watch-only – public keys only, cannot spend
```

The reconstructed descriptor is the genuine depth-2 shape `tr(Kint,{A,B},{C,D})` (four leaves, one tier each) -- the layout section 6.5 showed is cheaper to spend but that master cannot yet build. When a rust-miniscript release > 13.1.0 ships #953, this is rebuilt as a supported feature and this appendix retires.